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FINAL YEAR REPORT ON:

WAVELET AND ANN BASED FAULT DETECTION AND COMMUNICATION CONTROLLED SYSTEM

A final report submitted in partial fulfillment of the requirements for the degree of Bachelors of Electrical and Electronic Engineering

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ABSTRACT

This project presents an intelligent autonomous protection coordination system for local distribution power grids using Discrete Wavelet Transform (DWT) and Artificial Neural Networks (ANN) for rapid and accurate fault detection, classification, and isolation. Three-phase voltage and current signals obtained from the simulated power system are transformed into positive, negative, and zero sequence components to enhance fault identification capability. These signals are processed using DWT to extract transient features associated with various fault conditions. The extracted features are provided to an ANN-based classifier that identifies the operating condition of the system and determines the corresponding fault type. Based on the ANN decision, a control signal is generated to operate a circuit breaker, enabling automatic isolation of the faulted section and forming a closed-loop protection mechanism.

The complete system is modeled and simulated in MATLAB/Simulink, while real-time communication between the simulation environment and an Android-based monitoring application is achieved using Firebase Realtime Database. The Android application, developed using Android Studio in the Java programming language, displays system status and fault classification results in real time. The results demonstrate the effectiveness of the proposed framework in achieving intelligent fault detection, automated isolation, and real-time monitoring, thereby improving the reliability and operational efficiency of modern power distribution systems.

Keywords: Artificial Neural Network (ANN), Circuit Breaker Automation, Communication-Based Protection, Discrete Wavelet Transform (DWT), Fault Detection and Classification, Firebase Realtime Database, MATLAB/Simulink, Real-Time Monitoring, Sequence Components, Smart Grid Protection.

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CHAPTER 1: INTRODUCTION

1.1 Background

With the rise in demand for electrical power, distribution systems are increasingly subjected to heavy loading conditions. Transmission line fault detection plays a crucial role in maintaining the reliability and stability of electrical power systems. With the growing complexity and interconnectivity of modern grids, traditional fault detection methods face significant challenges in terms of accuracy, speed, and adaptability to dynamic operating conditions.

Protection schemes for local distribution grids that incorporate wavelet-based analysis and communication-assisted control address the challenges introduced by bidirectional power flows from distributed energy resources (DERs), such as solar and wind systems. Conventional protection approaches based on overcurrent relays and impedance measurements encounter limitations under dynamic fault profiles, high-impedance faults, and coordination issues caused by DER integration.

Wavelet Transform (WT) techniques provide improved fault analysis through time-frequency decomposition and multi-resolution analysis of transient signals. In this project, Discrete Wavelet Transform (DWT) is applied to extract transient features from voltage and current signals. These features are then used to train an Artificial Neural Network (ANN) model for intelligent fault detection and classification. The implemented system achieves high classification accuracy and enables automated circuit breaker operation through a closed-loop protection mechanism.

1.1.1 Wavelet Transform (WT) and Discrete Wavelet Transform (DWT)

Wavelet Transform is a signal-processing technique that decomposes signals into multiple scales, making it highly effective for analyzing non-stationary and transient signals in power systems. Traditionally, wavelet methods have been widely used for detecting, analyzing, and mitigating disturbances in transmission and distribution networks.

The Discrete Wavelet Transform (DWT) is a computationally efficient implementation of the wavelet transform that decomposes signals into discrete frequency bands and time shifts. Compared to Continuous Wavelet Transform (CWT), which provides

detailed frequency representation but requires higher computational resources, DWT is more suitable for real-time fault analysis and feature extraction applications.

1.1.2 Daubechies wavelets

Daubechies wavelets (dbN) are core tools in DWT for representing signals at multiple scales with good localization and smoothness. They are widely adopted due to their efficiency, mathematical strength, and broad applicability. Choosing the right dbN depends on your tradeoff between speed and accuracy. The simplest form, db1, is equivalent to the Haar wavelet and is fast but not smooth. As N increases (e.g., db2, db4, db10), the wavelets become smoother and better at capturing signal details, though they also require longer filters and more computation. These wavelets are particularly valued for their ability to perform multiresolution analysis, making them ideal for tasks like denoising, compression, and feature extraction. Daubechies wavelets are asymmetric by design, but related families like Symlets and Coiflets offer variations with improved symmetry or additional vanishing moments in the scaling functions.

Tools like ANN (Artificial Neural Network), MATLAB (Matrix Laboratory), helps in signal processing, wavelet analysis while modeling the protection coordination system that could be efficient for real time fault simulation and automation validation increasing the efficiency and sustainability of a protection system along with the transmission and distribution lines.

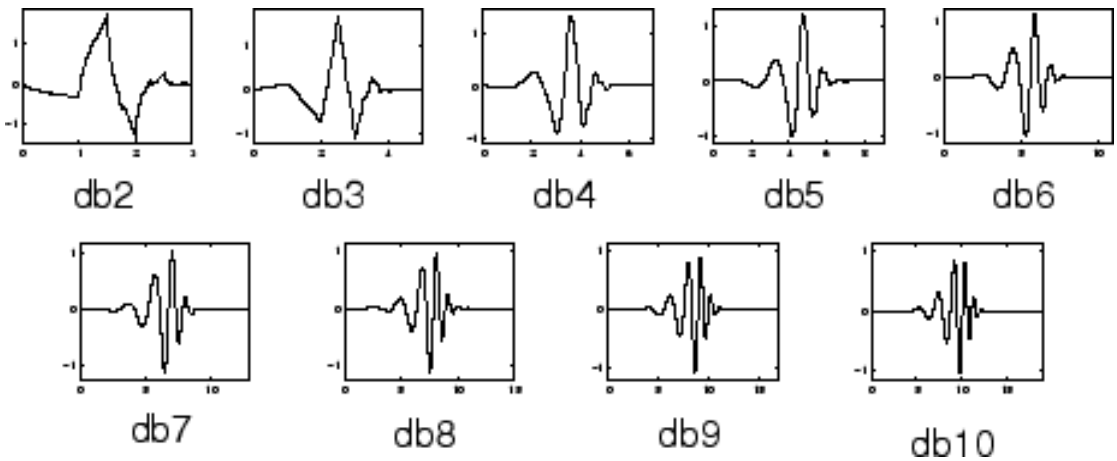

Figure 1.1: Daubechies Wavelet Families (db2 to db10)

1.1.3 Artificial Neural Networks (ANN)

Artificial Neural Networks (ANN) are computational models inspired by the human brain's neural structure. They are capable of learning from data, identifying complex patterns, and making intelligent decisions. ANNs consist of interconnected layers of neurons that process inputs through weighted connections, allowing them to adaptively improve performance over time.

In this project, ANN is employed as a core component for classifying and locating faults in power transmission lines. By training the ANN on features extracted using Discrete Wavelet Transform (DWT), the model learns to recognize the unique signatures of various fault types. This intelligent approach enables faster, and more accurate fault detection and localization compared to traditional methods, enhancing the reliability and responsiveness of power system protection.

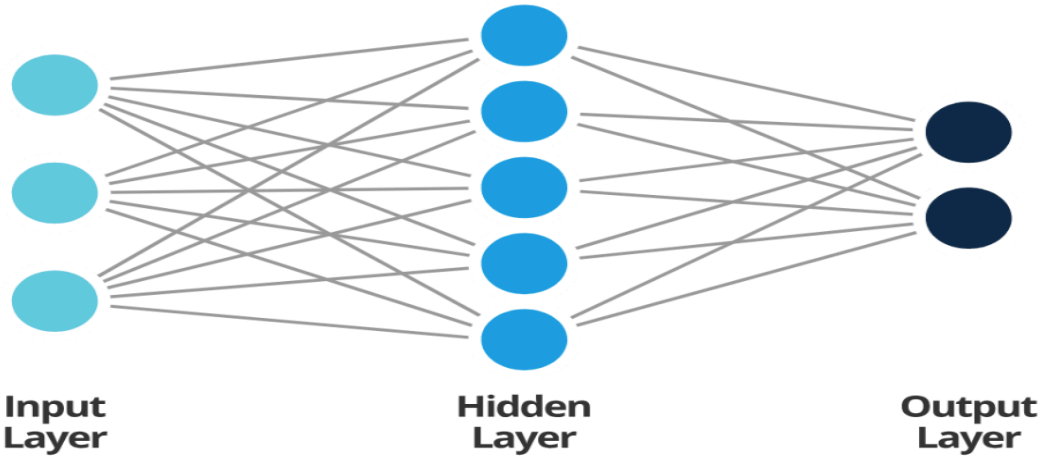

Figure 1.2 : Artificial Neural Network Architecture

1.2 Statement of the problem

Traditional fault detection, isolation, and service restoration (FDIR) in Nepal's power systems or distribution networks has historically been done with minimal automation, especially in rural areas. Fault detection is achieved by monitoring the energy of wavelet detail coefficients, while isolation is done by comparing these coefficients from different parts of the network to pinpoint the fault location. Although this method offers accurate time-frequency analysis, its implementation in Nepal is mostly restricted to urban or pilot systems, with rural areas still relying on manual fault isolation and service restoration by field staff.

Hence, the transmission and distribution systems in Nepal calls for an innovation using the recent developed technologies minimizing the effects, increasing efficiency while

reducing the manual labor and minimizing the costs of operation to maintenance. The proposed method of Fault Location, Isolation and Service Restoration (FLISR) uses the wavelet transform specially Daubechies wavelet family (db1to db5) as per the system designing to accurately detect and locate faults using real-time current and voltage signals captured by wavelet transform with communication support to make the system more promising and understandable.

1.3 Research gap

In recent years, considerable research has focused on fault detection in power systems, particularly in transmission networks. However, the application of intelligent protection strategies in local distribution grids, especially those that leverage real-time signal analysis and automation, remains limited. Most traditional protection systems rely on static thresholds and lack the ability to adapt to dynamic and evolving grid conditions, often leading to delayed or inaccurate fault isolation.

While wavelet transform techniques have proven effective in extracting high-frequency transient features associated with faults, and Artificial Neural Networks (ANNs) excel in classification tasks, their combined potential remains underutilized in practical distribution grid protection schemes. Furthermore, current studies often omit the integration of real-time control logic and communication frameworks that could enable automated and intelligent circuit breaker operations based on ANN outputs.

Existing simulation approaches tend to emphasize component-level modeling but fail to implement a closed-loop protection system that includes signal acquisition, feature extraction, intelligent fault classification, and communication-based control actions. The lack of such an end-to-end approach restricts the practical implementation of intelligent fault management systems in localized grids, especially in developing or semi-urban regions.

This project aims to bridge these gaps by developing a smart protection coordination scheme that uses Discrete Wavelet Transform (DWT) for feature extraction and a trained ANN model for fault detection and classification. The ANN output will directly control circuit breaker actions via a communication interface, ensuring real-time fault isolation and restoration. By simulating and validating this setup in a localized grid

environment, the study targets not only improved accuracy and speed of protection but also the automation and resilience required for modern smart grids.

1.4 Objectives

1.4.1 Main objectives

To design, develop, and validate an intelligent fault detection, classification, isolation, and service restoration system for transmission lines using Discrete Wavelet Transform (DWT) and Artificial Neural Networks (ANN), enabling fast protective action and improved overall power system reliability.

1.4.2 Specific objectives

I. To design and develop a real-time intelligent fault classification system for power transmission lines using wavelet-based signal processing techniques combined with Artificial Neural Networks (ANN).

II. To optimize the ANN architecture to achieve high fault detection and classification accuracy while minimizing misclassification through comprehensive training using wavelet-extracted features.

III. To simulate various fault conditions in MATLAB Simulink and evaluate the performance and robustness of the proposed ANN–DWT model using synthetic fault datasets.

IV. To evaluate and compare the proposed approach with conventional fault detection and classification methods in terms of detection accuracy, response time, and computational latency.

V. To design and implement an Android-based monitoring and communication application using Firebase for real-time visualization of fault status and protective system operation.

1.5 Scopes of the project

- Improved Fault Detection:
  This project attempts to improve the precision, responsiveness, and detail level of the fault detection process in power transmission lines. The system would allow for rapid identification and precise fault pinpointing to reduce system outages.

- Real Time Monitoring and Predictive Analytics:
  Integration of ANN-based models allows the system to perform continuous real-time analysis of the data from the transmission line and assists in proactive fault detection and maintenance scheduling.
- Advanced Fault Type Classification:
  By deploying data-driven deep learning algorithms, this solution can asymmetrically classify multi-dimensional advanced faults to improve diagnostics and enhance operational decisions.
- Accelerated Service Restoration:
  The system's ability to quickly detect and classify faults enables faster service restoration, leading to improved grid reliability, resilience, and reduced MTTR.
- Economic and Operational Efficiency:
  Advanced fault diagnostics enable faster response times which saves costs by preventing equipment damage, energy waste, unnecessary manual inspections, and optimizing overall system performance.

1.6 Limitations of the project

- Simulated Data Dependency
  The training is carried out using simulated fault data which does not cover all aspects of variability, affecting model generalization.
- Data and Resource Limitations
  Restricted access to real-life data can significantly impact the performance of ANN models, which require exhaustive and varied datasets, as well as ample computing resources.
- Difficulties in Real-Time Deployment
  Integration and latency constraints, as well as processor limitations, pose challenges in the deployment of ANN models onto live systems.
- Limitations in Scalability
  The precision and accuracy of the model is likely to deteriorate with an increase in the size and complexity of grids and multiple concurrent faults.

CHAPTER 2: LITERATURE REVIEW

Fault detection, service restoration and classification in power systems have significantly advanced with the integration of wavelet and artificial neural networks. Key studies include:

Research papers:

A. Fault detection, classification and service restoration in power systems are critical for maintaining stability and reliability. Authors in [1]S.A Gafoor and P.V. Rao presented a fault diagnosis, classification, and location method for transmission lines based on wavelet transforms and artificial neural networks (ANNs). Synchronized sequences of voltage and current signals were conducted through Bior2.2 wavelets a determine the detection or classification fault index, while Bior4.4 wavelets are undertaken to take features for estimating location. Then two different ANNs are trained: i.e. ground fault ANN and phase fault ANN, their backbone of training involves estimating critical fault classes quickly and accurately and with respect to fault contacts to performance.

B. S. Senthilmurugan at [2] focused on load sharing and power re-routing during fault conditions explaining the Fault Locations, Isolations and Service Restoration (FLISR). The overall simulation was achieved using MATLAB Simulink in standard 9 bus electrical network where initially load flow analysis and fault analysis were studied with the system undergoing three phases to ground fault current. After which the protection and rerouting solutions were infused to the existing IEEE-9 BUS and results were achieved. This paper elaborated the key feature id self-healing in the aspect of power distribution and uninterrupted power supply to the load.

C. Arun Rajendra at [3] presented a novel methodology for finding fault location by using Wavelet Transform (WT) analysis and isolating respective faulted zone from rest of the Low Voltage Direct Current (LVDC) microgrid so as to prevent entire system failure implemented using ring bus topology. Here the PPU (Power Probe Unit) circuit was loaded into microcontroller circuit and wavelet analysis of the probe voltage response was done by microcontroller. By

using the extracted frequency, the distance at which fault occurred was measured and displayed on the LCD display. Implementation of the system with PPU circuit is done using the MATLAB and PROTEUS simulation software.

D. Ranjit at [4]described a fault location estimation technique for transmission line based on the Discrete Wavelet Transform (DWT) based on wavelet-scale-based signal processing in order to extract fault data from current signals and a decision tree classifier for more accurate estimation. Classification tree was used to train and determine the fault location and classify the type of a standard IEEE-9 bus system and the same algorithm was implemented in Integrated Nepal Power System with an approximate error of less than 0.5%.

E. R. V. R at [5] proposed a smart grid protection coordination scheme that integrates communication-assisted recloser coordination with wavelet-based fault identification to increase distribution system reliability. Communication-enabled reclosers (GPRS or indirect sensing) could coordinate with one another to trip equipment to isolate only the faulty section or meter, avoiding unnecessary upstream tripping, while high-frequency travelling waves are examined through application of Discrete Wavelet Transform (DWT) to identify fault location when a fault does occur. In the case of an upstream fault, a load transfer logic (TP's logic) was used to connect and switch vital loads onto healthy feeders. The use of fast fault localization, and adaptive protection settings in the PSCAD/EMTDCTM simulations demonstrated decrease in restoration periods, from hours to seconds. The scheme mitigated adverse outage implications to improve overall grid reliability, increases grid resiliency, and assists worker safety when crews were involved to restore power.

F. S Voima at [6] proposed an adaptive protection scheme for Smart Grids to mitigate protection challenges associated with distributed generation (DG), including issues of bidirectional power flow and low fault currents. Protection using directional overcurrent and distance protection was considered. The overcurrent protection scheme performed adequately during grid-connected operation but bemusedly failed during the islanded operation, and only for the fault current to the right, where the untapped DG elements could not compensate

for the insufficient fault current delivered. Based on the network local conditions monitored in real time, the adaptive protection scheme had the ability to automatically change protection settings in either a centralized or decentralized approach. Selectivity and speed were assured using communication between the relays. Simulation results confirmed each money was used effectively while still also providing seamless transfer between the islanded and grid-connected operation while preserving strong fault detection even when there was only 10-degree phase shift fault current to the left.

G. K. Prasanna at [7] proposed a single-ended traveling wave-based fault location technique aimed at hybrid transmission lines (overhead + underground cable) via Discrete Wavelet Transform (DWT) and Support Vector Machines (SVM). The normalized wavelet energies were used as inputs to SVM classifiers, which identified first the faulty section (overhead line or cable) and then faults that resided within either of the first or second halves. For fault location, time delays (Δt) at each peak of the travelling waves from the aerial mode voltage wavelet coefficients, were used with the respective known wave propagation velocities in both media. The methodology was fault-type independent, can manage high-impedance and non-ideal faults, and exceeds >98% accuracy in determining only the section of the fault with errors in the location estimation being less than 2%. It had been successfully validated in the form of ATP simulations using MATLAB to process the transients before performing the fault location and was effective in overcoming issues that compromise the fault location approaches for smart grids incorporating hybrid line topologies.

H. Srivasta at [8] presents a machine learning-based framework for fault isolation and fault location prediction in a meshed AC microgrid. The framework used post-fault voltage and current measurements from the source buses to train two modules, namely; a) Fault Isolation Module (FIM), and b) Fault Locator Module (FLM). The FIM used Support Vector Machine (SVM) to classify the faulted line and fault type, while the FLM used Gaussian Process Regression (GPR) to predict the true location of the fault. The methodology was validated with a 14-bus microgrid model that varied load and DG penetration levels. The results suggest that SVM delivers 99.5% accuracy for fault isolation, and that

GPR outperforms the other methods to distinguish fault location with little error. The framework establishes rapid and reliable protection of the microgrid after fault detection and clearance.

I. Omer at [9] proposed a machine learning method (ML) for detecting, classifying, and locating faults in 400 kV high-voltage transmission line that together use a three-stage cascade method to improve accuracy. The first stage was to detect the fault using an ensemble bagged trees algorithm, achieving 100% accuracy to detect fault from normal. The second stage classified the fault with a trilayered neural net to identify the faulty phase or phases with 99.97% accuracy. The fault location diagnostics externalized with ML regression methods, using interaction linear regression for phase-to-phase faults and the stepwise linear regression for phase-to-ground faults, with errors at maximum of 110 m and 26 m, respectively, over a 178 km transmission line. The method was contingent on simulated electrical parameters generated from actual line characteristics.

J. El Safty at [10] present an on-line fault detection scheme for transmission lines using Artificial Neural Networks (ANNs). In their work, the authors employed post-fault voltage and current waveforms obtained from a simulated power system modeled in PSCAD/EMTDC and from an experimental transmission line setup. To enhance fault discrimination and reduce ANN complexity, the voltage and current signals were preprocessed using a voltage–current (V–I) subtraction technique followed by the Sum of Averaged Difference (SADI) feature extraction method. A multilayer feed-forward ANN trained using the backpropagation algorithm was then developed to identify faulted phases and ground involvement. The proposed approach was validated under various fault conditions, including line-to-ground, line-to-line, double line-to-ground, and three-phase faults. Experimental implementation using signal conditioning circuits and a data acquisition system demonstrated correct fault detection and tripping action in real time. The results confirm that the ANN-based method is simple, effective, and suitable for practical on-line transmission line protection application.

K. Tayeb at [11] proposed an Artificial Neural Network (ANN)–based scheme for fault detection, classification, and location in power transmission lines. The study employed backpropagation neural networks to implement a complete distance protection system by dividing the transmission line into multiple protection zones. A 110-kV overhead transmission line was modeled using a π-section representation in MATLAB/Simulink, and three-phase voltage and current measurements at the sending end were used as ANN inputs. Separate ANN architectures were developed for fault detection, fault type classification, and fault zone identification under single line-to-ground, line-to-line, and double line-to-ground fault conditions. Simulation results demonstrated fast response and high accuracy, confirming the effectiveness of ANN-based protection for intelligent transmission line fault diagnosis.

L. Özüpak at [12] presents a machine learning-based comparative framework for transmission line fault detection and classification in high-voltage power systems. The study utilizes three supervised learning algorithms, namely Support Vector Machines (SVM), Logistic Regression (LR), and Decision Tree (DT), with random search hyperparameter optimization to enhance model performance. Instantaneous three-phase voltage and current signals obtained from a 154 kV MATLAB/Simulink-based transmission line model were used to generate labeled datasets under faulted and fault-free conditions. The dataset included multiple fault types such as line-to-ground, line-to-line, double-line-to-ground, and three-phase faults. The models were trained using an 80–20 train–test split and evaluated using accuracy, RMSE, MAE, and R^2 metrics. The results demonstrate that the Decision Tree model achieved superior performance with 99.87% test accuracy, significantly outperforming Logistic Regression (97.01%) and SVM (74.19%). The findings confirm that decision tree-based models, when optimized with random search, provide fast, accurate, and reliable fault detection and classification, making them highly suitable for real-time transmission line protection applications.

M. Hammed at [13] present a hybrid Wavelet Transform–Artificial Neural Network (WT–ANN) based framework for fault detection and classification in power distribution systems. The proposed approach utilized post-fault voltage

and current signals, which were decomposed using discrete wavelet transform to extract transient fault features across multiple frequency bands. These wavelet-based features were then supplied to an ANN classifier for identifying faulted conditions and fault types. The methodology was validated through simulation studies under various fault scenarios, including different fault types and fault resistances. The results indicate that the WT–ANN model achieves high classification accuracy and strong noise immunity compared to conventional techniques. The study confirms that the integration of wavelet-based feature extraction with ANN enhances reliability, speed, and robustness of fault diagnosis, making it suitable for real-time protection applications in distribution networks.

N. Alhanaf *at* [14] proposed intelligent fault detection, classification, and location schemes for smart grids using deep neural networks. In their work, the authors developed data-driven models based on Artificial Neural Networks (ANNs) and one-dimensional Convolutional Neural Networks (1D-CNNs) using raw three-phase voltage and current measurements without feature engineering or signal transformation. The proposed framework simultaneously identified the faulty line, classified fault types, and estimated fault location in an IEEE 6-bus power system under various fault scenarios, including symmetrical and unsymmetrical faults with different fault resistances and locations. Extensive simulations conducted in MATLAB/Simulink demonstrated high performance, achieving accuracy rates of up to 99.99% for faulted line identification, 99.99% for fault classification, and above 96% for fault location. The results confirm that deep learning-based approaches significantly enhance the speed, accuracy, and reliability of fault diagnosis in smart grids, making them well suited for real-time protection and self-healing power system applications.

O. Pancha at [15]presented a Wavelet Transform and Artificial Neural Network (WT–ANN) based islanding detection scheme for a distributed generation integrated distribution feeder. Negative sequence voltage signals at the point of common coupling were analyzed using discrete wavelet transform to extract transient features, which were then classified using a backpropagation-based

ANN. The method achieved high classification accuracy and effectively eliminated the non-detection zone without injecting disturbances, ensuring power quality preservation. This work directly supports the proposed final-year project by demonstrating the effectiveness of wavelet-based feature extraction combined with ANN classification for detecting abnormal power system events. The methodology can be extended in the proposed project for fast fault detection and classification, while the decision outputs can be integrated with a communication-controlled protection system for real-time monitoring, remote tripping, and intelligent grid automation.

P. Chen at [16] proposed an Artificial Neural Network (ANN)–based fault diagnosis model for distribution networks to achieve fast and accurate identification of different fault types. In their work, transient voltage and current data obtained from fault indicators were first processed using Principal Component Analysis (PCA) to reduce data dimensionality and extract dominant fault features. The reduced feature set was then used to train a backpropagation-based ANN to classify single-phase grounding, short-circuit, and multi-phase grounding faults. Simulation results showed that the proposed ANN model achieved an overall fault identification accuracy of approximately 93.2% and outperformed traditional decision tree and support vector machine methods. This study supports the proposed final-year project by validating ANN as an effective classifier for transient fault data and highlighting the importance of feature extraction prior to classification. The methodology can be extended in the proposed project by replacing PCA with Wavelet Transform for improved time–frequency feature extraction and integrating the ANN output with a communication-controlled protection system for real-time fault reporting and remote-control actions.

Q. Mrs. S. Kirubadevi at [17] proposed a wavelet transform–based fault identification and classification method for high-voltage transmission lines to ensure fast and reliable power system protection. In their work, three-phase current signals obtained at the relay location during normal and faulty conditions were analyzed using the Discrete Wavelet Transform (DWT) with the Daubechies-8 (db8) wavelet to capture transient fault characteristics. Fault

detection was achieved by comparing the norm of wavelet coefficients with predefined threshold values, while fault classification was performed using the summation of multi-level wavelet decomposition coefficients to identify the involved phases and fault type. The method successfully classified single line-to-ground, line-to-line, double line-to-ground, and three-phase faults through MATLAB/Simulink simulations. Results demonstrated that wavelet-based analysis provides accurate time–frequency localization of fault transients and is well suited for transmission line protection. This study supports the proposed final-year project by validating wavelet transform as an effective feature extraction technique for transient fault signals and can be further extended by integrating intelligent classifiers such as ANN and communication-based protection schemes for real-time fault monitoring and control.

Books:

1) Modern Power System Analysis by D. P. Kothari and I. J. Nagrath [18] provides foundational insights into the analysis, operation, and optimization of power systems. It covers fault analysis, system stability, and protection mechanisms crucial for fault detection and location.
2) Electrical Power Systems by Ashfaq Hussain [19] offers a comprehensive understanding of power system design, including fault analysis methods. It explains protective systems and algorithms, forming a theoretical basis for modern techniques.
3) Fundamentals of Power system protection by Y.G. Paithanker and S.R.Bhide b [20] offers a detailed understanding of relays and power system protection work mechanisms.

2.1 Summary of Literature Review

An in-depth review of existing literature establishes that the integration of wavelet-based signal processing with intelligent learning techniques provides a reliable and effective foundation for advanced fault management in power distribution systems. Numerous studies demonstrate that wavelet transforms offer superior time–frequency localization, enabling precise extraction of transient fault features from voltage and current signals even under noisy and non-stationary operating conditions. When these

wavelet-derived features are processed using artificial neural networks and other machine learning models, fault detection and classification accuracy are significantly improved, while decision-making speed is enhanced compared to conventional relay-based protection schemes. Comparative analyses further indicate that optimized learning algorithms such as ANN, decision trees, and support vector machines exhibit strong adaptability and generalization capability across various fault types, loading conditions, and network configurations, especially when combined with hybrid frameworks and hyperparameter optimization. However, existing research predominantly focuses on isolated tasks such as fault detection or classification, with limited emphasis on coordinated fault location, automated isolation, and integrated service restoration. This identified gap directly motivates the present work, which proposes a comprehensive Wavelet–ANN-based intelligent protection and control framework aimed at achieving accurate fault localization, effective isolation, and faster service restoration, thereby improving reliability and operational resilience in modern distribution networks.

CHAPTER 3: METHODOLOGY

3.1 Block Diagram

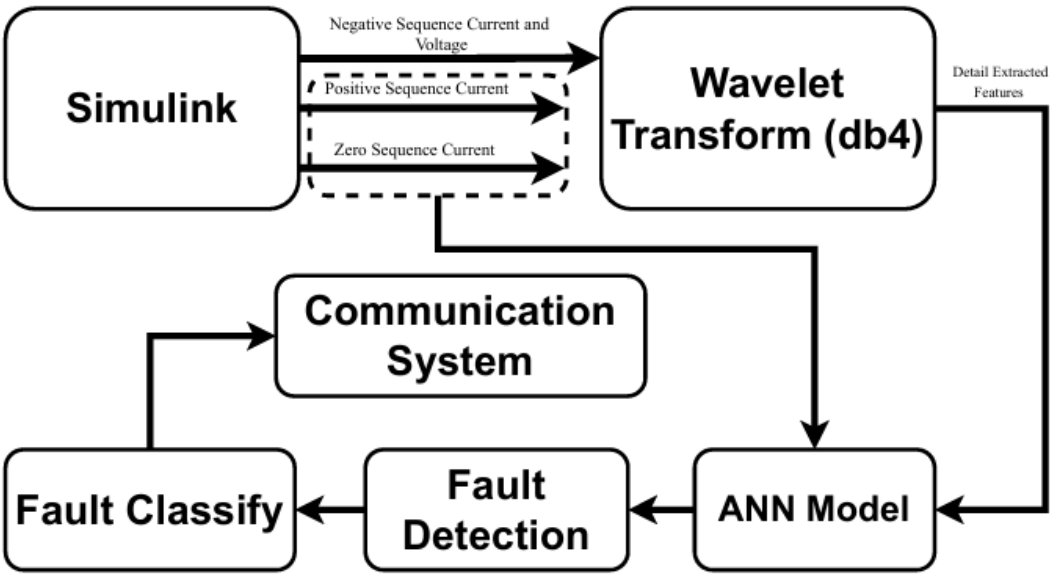

Figure 3.1: Block Diagram

The proposed intelligent fault detection and classification system follows a structured signal processing and decision-making framework, as illustrated in the methodology block diagram. The overall process integrates power system simulation, wavelet-based feature extraction, artificial neural network-based classification, and communication-assisted monitoring.

1. Power System Simulation in MATLAB Simulink:

The methodology begins with the development of a power transmission system model in MATLAB Simulink. The simulated system is subjected to both normal and faulty operating conditions. Voltage and current signals are measured from the transmission line, and sequence components positive, negative, and zero sequence currents and voltages are extracted. These sequence components enhance sensitivity to fault conditions and improve discrimination between different fault types.

2. Wavelet Transform-Based Feature Extraction

The extracted sequence signals are processed using the Discrete Wavelet Transform (DWT) with the Daubechies-4 (db4) wavelet. DWT provides effective time–frequency analysis, enabling the capture of transient characteristics introduced during fault events. Multi-level decomposition is performed, and the resulting wavelet detail coefficients

represent the high-frequency components associated with fault inception. These coefficients form the feature set used for intelligent decision-making.

3. Artificial Neural Network (ANN) Model

The wavelet-extracted features are supplied as inputs to the Artificial Neural Network model. The ANN is trained using supervised learning techniques with datasets generated from various fault scenarios. Through training, the network learns the nonlinear relationship between wavelet features and corresponding fault classes. Once trained, the ANN performs real-time fault detection and classification based on incoming signal features.

4. Fault Detection and Classification

The ANN output is processed to determine the presence of a fault and to identify the fault type. This intelligent classification approach enables fast and accurate decision-making compared to conventional threshold-based protection methods. The classification results are further used to initiate appropriate protective actions.

5. Communication System Integration

A communication system is integrated to transmit the ANN decision outputs to external monitoring and control interfaces. This communication framework enables real-time sharing of fault status and system conditions. In the implemented system, Firebase is used as a cloud-based communication platform to synchronize data between the ANN model and the Android application.

6. Android-Based Monitoring Interface

The final stage of the methodology includes an Android application developed using Android Studio. The application retrieves fault detection and classification information from Firebase and displays the real-time system status to the user. This provides remote monitoring capability and enhances situational awareness during fault events.

3.2 Workflow

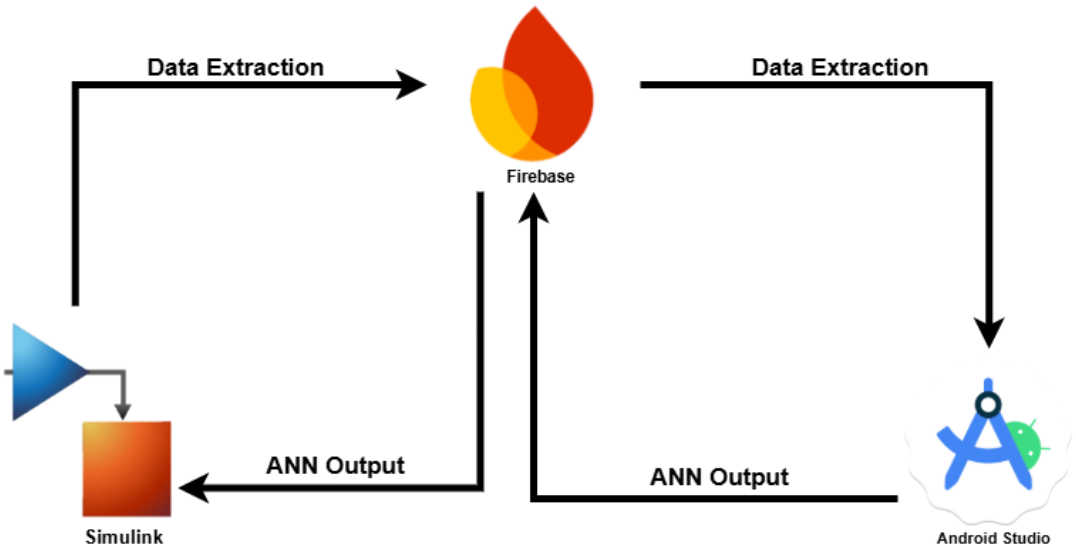

Figure 3.2: Workflow

The proposed workflow begins with the acquisition of voltage and current signals from the MATLAB Simulink–based power transmission system under both normal and faulty conditions. Along with the raw signals, positive, negative, and zero sequence components are extracted to enhance fault sensitivity and improve discrimination between different fault types. These signals are then processed using wavelet-based feature extraction techniques to capture transient fault characteristics. The extracted features are analyzed using an Artificial Neural Network (ANN) for intelligent fault detection and classification. The ANN decision outputs are transmitted through a cloud-based communication framework using Firebase and are finally visualized in real time through an Android application developed using Android Studio.

3.2.1 Data Extraction

The data extraction process begins with the acquisition of three-phase voltage and current signals from the power transmission system modeled in MATLAB Simulink. These signals are continuously monitored under both normal and faulted operating conditions. To enhance fault sensitivity and enable effective discrimination between fault types, the acquired voltage and current signals are converted into their corresponding positive, negative, and zero sequence components within the Simulink environment using sequence component transformation blocks.

The extracted sequence components are then processed using the Discrete Wavelet Transform (DWT) with the Daubechies-4 (db4) wavelet. Wavelet decomposition is performed to capture transient and high-frequency components associated with fault events. The resulting wavelet detail coefficients represent compact and informative features that characterize system behavior during normal and faulty conditions.

All extracted features, including voltage and current sequence data and wavelet coefficients, are transmitted in real time to a cloud-based communication platform using Firebase. This real-time data transfer is achieved through MATLAB scripts that interface with Firebase, enabling continuous synchronization between the simulation environment and external applications.

An Android application developed using Android Studio and programmed in Java retrieves the uploaded data from Firebase. The application includes an integrated Artificial Neural Network (ANN) model, which processes the incoming wavelet-based features to perform intelligent fault detection and classification. The final decision results, including system status and fault type, are displayed in real time on the mobile application, providing an effective remote monitoring and visualization platform.

3.2.1.1 Model Simulation

A three-bus 50 Hz power system is modeled in MATLAB Simulink with Bus 1 and Bus 4 as generator buses and Bus 2 , Bus 3 as a load bus. The buses are connected through two 11 kV, 100 km transmission lines, with static loads of 10 MW, 15 MW, and 5 MW connected at Buses 1, 2, and 3 respectively.

Voltage and current measurements are obtained and converted into positive, negative, and zero sequence components for wavelet-based feature extraction and ANN classification. No fault, AG, BG, CG, AB, AC, BC, ABC, ABCG, ABG, ACG and BCG faults with 50 Ω fault resistance are simulated.

Circuit breakers are integrated and controlled through a communication-assisted closed-loop mechanism. The ANN-based fault classification result is transmitted via Firebase to MATLAB, where it is used to generate control signals for automatic breaker tripping, enabling intelligent real-time fault isolation.

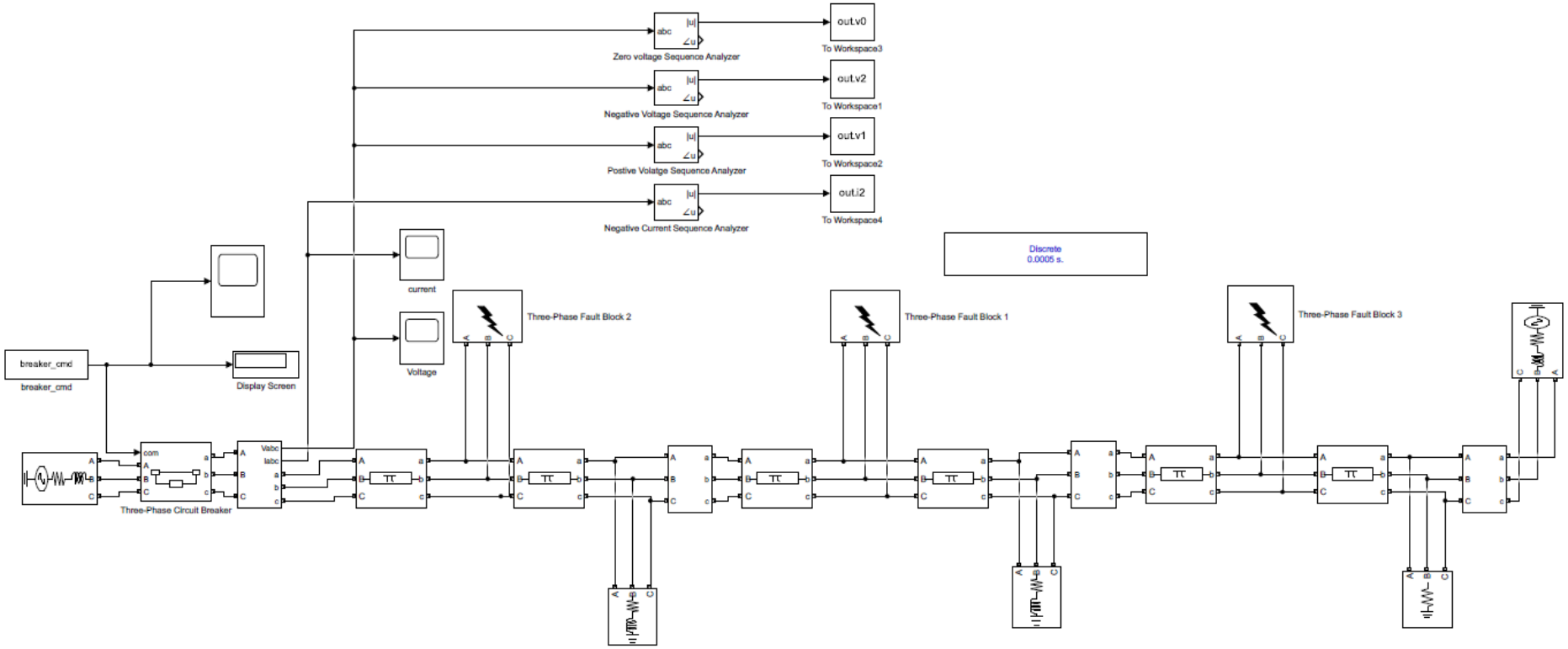

Figure 3.3 : Simulation Model

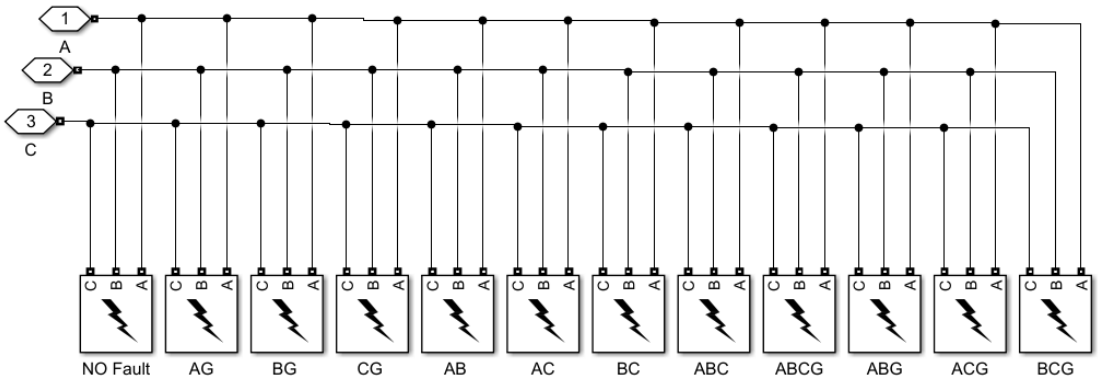

Figure 3.4:Subsystem of all types of faults

3.2.1 Real-Time Data Transfer to Android Application Using Firebase Realtime Database

After feature extraction, the extracted voltage, current, and sequence component data are transmitted in real time from the MATLAB environment to a cloud-based communication platform. Firebase Realtime Database is used as the communication interface to enable continuous and low-latency data synchronization between MATLAB and the Android application.

A MATLAB script is employed to establish a real-time connection with the Firebase Realtime Database. The processed signal features are uploaded in structured JSON format whenever a change in system condition is detected. Firebase automatically synchronizes the updated data across all connected clients, ensuring timely availability of system information.

An Android application developed using Android Studio in the Java programming language retrieves the real-time data from Firebase. The application continuously monitors database updates and forwards the received data to the integrated ANN model for intelligent fault detection and classification. This integration enables real-time remote monitoring and communication-assisted protection of the power system.

3.2.2 Android-Based Application with Integrated ANN Model

The Android application is developed using Android Studio with Java as the programming language and serves as the final decision-making and monitoring

interface of the proposed system. The application retrieves real-time data from the Firebase Realtime Database and processes the incoming wavelet-based features using an integrated Artificial Neural Network (ANN) model.

The ANN embedded within the application performs intelligent fault detection and classification based on the received data. By executing the ANN directly on the mobile platform, the system enables fast decision-making without reliance on continuous external computation. The classification results and system status are displayed in real time through a user-friendly interface.

This integration of ANN within the Android application demonstrates the feasibility of deploying intelligent protection algorithms on mobile platforms, enabling remote monitoring and communication-assisted protection in smart grid applications.

3.3 Artificial Neural Network (ANN) Implementation

An Artificial Neural Network (ANN) is employed as the intelligent decision-making component of the proposed fault detection and classification system. The ANN processes wavelet-based and sequence-component features extracted from voltage and current signals and determines the corresponding fault category. The ANN model is trained offline using labeled datasets and subsequently integrated into the Android application for real-time fault classification.

3.3.1 Dataset Preparation and Feature Processing

The dataset used for ANN training is generated from the MATLAB Simulink–based power system model under normal and various faulted operating conditions. Voltage and current signals are processed to obtain positive, negative, and zero sequence components. Additionally, Discrete Wavelet Transform (DWT) is applied to voltage and current signals to extract transient fault features.

The input feature vector to the ANN consists of twelve features, including:

- Positive Sequence Voltage
- Negative Sequence Voltage
- Zero Sequence Voltage
- Negative Sequence Current

- Voltage wavelet detail coefficients: D1_v2, D2_v2, D3_v2, D4_v2
- Current wavelet detail coefficients: D1_i2, D2_i2, D3_i2, D4_i2

These features are normalized using standardization to ensure zero mean and unit variance, which improves convergence during ANN training. Fault labels are encoded into numerical form to support multi-class classification. The dataset is divided into training and validation sets using an 80:20 split while maintaining class balance. This ensures reliable performance evaluation during training.

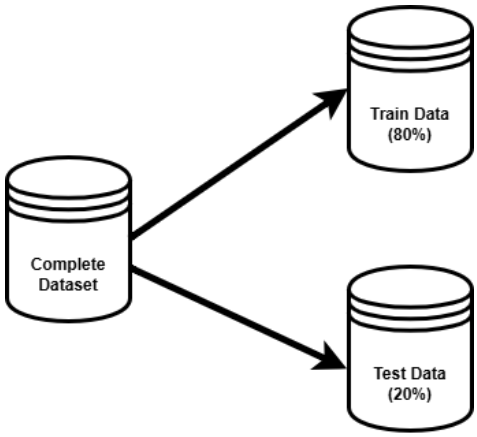

Figure 3.5:Data Separation

3.3.2 ANN Architecture

The implemented ANN follows a Multilayer Perceptron (MLP) architecture comprising an input layer, multiple hidden layers, and an output layer, as illustrated in Figure 3.6.

The input layer consists of twelve neurons corresponding to the extracted electrical and wavelet-based features. These inputs provide both steady-state and transient information, enhancing fault discrimination capability.

The hidden layer section includes four fully connected layers with 512, 256, 128, and 64 neurons, respectively. Each hidden layer uses the Rectified Linear Unit (ReLU) activation function to introduce nonlinearity. Dropout regularization is applied to reduce overfitting and improve generalization performance.

The input feature vector to the ANN consists of twelve features, including:

- Positive Sequence Voltage
- Negative Sequence Voltage

- Zero Sequence Voltage
- Negative Sequence Current
- Voltage wavelet detail coefficients: D1_v2, D2_v2, D3_v2, D4_v2
- Current wavelet detail coefficients: D1_i2, D2_i2, D3_i2, D4_i2

The output layer contains twelve neurons, each representing a specific fault class. The considered fault classes include:

- Single line-to-ground faults: AG, BG, CG
- Line-to-line faults: AB, BC, AC
- Double line-to-ground faults: ABG, BCG, ACG
- Three-phase fault: ABC
- Three-phase-to-ground fault: ABCG
- No - Fault

The output layer produces class scores, and the final fault classification is obtained using a Softmax-based decision through the cross-entropy loss function during training.

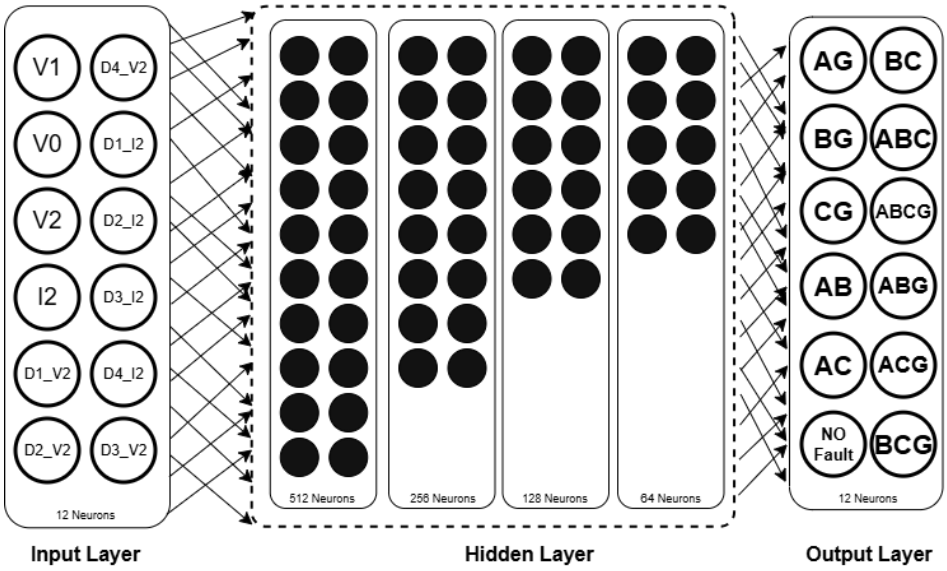

Figure 3.6: ANN Architecture

3.3.3 Training Process

The ANN is trained using supervised learning with labeled datasets. The training process is implemented using the PyTorch framework. The dataset is divided into training and validation sets using an 80:20 split while preserving class distribution.

The Adam optimization algorithm is employed with a learning rate of 0.001, and cross-entropy loss is used as the objective function to handle multi-class classification. Training is performed for 120 epochs with a batch size of 512. During training, validation accuracy is monitored to evaluate model performance and ensure stable learning.

To ensure reproducibility, fixed random seeds are used, and deterministic training behavior is enforced.

3.3.4 Model Saving and Deployment

After training, the ANN model parameters are saved along with essential preprocessing information, including feature normalization parameters, fault class labels, and feature names. These stored parameters ensure consistent preprocessing and classification during real-time operation.

The trained ANN is integrated into the Android application, where it processes real-time feature data retrieved from the Firebase Realtime Database. The ANN performs fault classification within the mobile application and provides immediate classification results for monitoring and protective decision support.

3.3.5 Role of ANN in the Proposed System

The integration of the ANN within the Android application enables decentralized and intelligent fault classification. By combining wavelet-based transient features with sequence component information, the ANN enhances classification accuracy and robustness. This approach demonstrates the feasibility of deploying intelligent protection algorithms on mobile platforms for real-time monitoring and communication-assisted power system protection.

3.4 Communication-Assisted Control and Circuit Breaker Operation

The proposed system incorporates a bidirectional communication framework to enable real-time fault detection, classification, and protective control. After processing the incoming data, the integrated Artificial Neural Network (ANN) within the Android application determines the fault type based on the received features. The classification result is then transmitted back to the cloud using the Firebase Realtime Database.

The Android application updates the Firebase database with the ANN output, including the detected fault category and system status. Firebase serves as a centralized communication platform that ensures low-latency and reliable data synchronization between the mobile application and the MATLAB environment.

A MATLAB script continuously monitors the Firebase Realtime Database and retrieves the updated fault classification data generated by the Android application. The received information is stored in the MATLAB workspace in real time. This workspace variable acts as a control signal for protective decision-making within the Simulink model.

Using the fault classification information stored in the MATLAB workspace, control logic is implemented to operate the circuit breaker in the Simulink model. When a fault is detected, the corresponding breaker is tripped to isolate the faulted section of the transmission system. This real-time feedback loop enables communication-assisted protection by linking intelligent fault classification with automated circuit breaker operation.

The closed-loop interaction between MATLAB Simulink, Firebase, and the Android application demonstrates a practical implementation of real-time intelligent protection and control. This approach enhances system responsiveness and reliability by integrating signal processing, intelligent decision-making, and communication-based control within a unified framework.

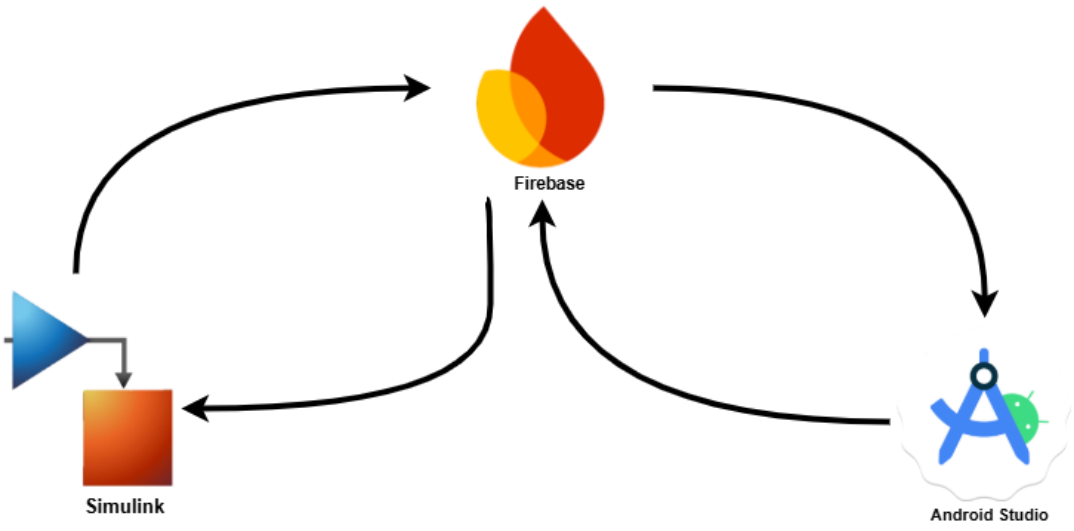

Figure 3.7: Communication system loop

3.5 Tools and Technologies Used

A. MATLAB and Simulink

MATLAB is a high-level programming platform widely used for numerical computation, data analysis, and algorithm development in engineering applications. Simulink, an integrated graphical modeling environment within MATLAB, provides block-based tools for modeling and simulating dynamic systems such as power networks.

In this project, MATLAB Simulink is used to model and simulate a local power distribution system under various operating conditions. Different fault scenarios are introduced into the system, including:

- Single Line-to-Ground (LG) faults
- Line-to-Line (LL) faults
- Double Line-to-Ground (LLG) faults
- Three-phase (LLL) faults

The simulation environment enables the generation of synthetic voltage and current signals that represent system behavior during both normal and faulted conditions. These signals serve as the primary data source for wavelet-based feature extraction and ANN training.

Simulink also supports:

Modeling of generators, transmission lines, loads, relays, and circuit breakers

Observation of fault transients and breaker operations

Implementation of control logic for protective actions

The seamless integration between MATLAB and Simulink allows efficient signal extraction, application of Discrete Wavelet Transform (DWT), real-time data handling, and post-simulation analysis. Together, MATLAB and Simulink provide a comprehensive framework for system modeling, signal processing, intelligent analysis, and protection logic implementation.

B. Discrete Wavelet Transform (DWT)

The Discrete Wavelet Transform (DWT) is employed to extract time–frequency information from voltage and current signals obtained from the simulated power system. Power system faults generate high-frequency transient components that are difficult to capture using conventional signal processing techniques.

In this work, the Daubechies-4 (db4) wavelet is used due to its suitability for transient analysis and its effectiveness in power system fault studies. Multi-level wavelet decomposition enables isolation of fault-related signal components.

- DWT plays a critical role in:
- Detecting transient disturbances caused by faults
- Extracting discriminative wavelet detail coefficients
- Reducing signal dimensionality while preserving fault characteristics

The wavelet-extracted features form a reliable input set for the Artificial Neural Network, significantly improving fault detection and classification performance.

C. Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is used as the intelligent decision-making component of the proposed system. The ANN performs fault detection and classification based on wavelet-extracted features and sequence component information derived from voltage and current signals.

A feedforward Multilayer Perceptron (MLP) architecture trained using supervised learning is implemented. The ANN learns the nonlinear relationship between the extracted feature set (inputs) and the corresponding fault categories (outputs).

The ANN is responsible for:

- Identifying the presence of a fault
- Classifying the type of fault (LG, LL, LLG, LLL, etc.)
- Supporting real-time intelligent protection decisions

The trained ANN model is deployed within the Android application to enable decentralized and real-time fault classification, demonstrating the feasibility of mobile-based intelligent protection systems.

D. Android Studio

Android Studio is the official Integrated Development Environment (IDE) for Android application development, provided by Google. It offers comprehensive tools for application design, development, debugging, and deployment.

In this project, Android Studio is used to develop a mobile-based monitoring and decision interface using the Java programming language. The Android application serves as the front-end platform for real-time visualization of system status and ANN-based fault classification results.

The application enables:

- Real-time display of fault classification results
- Visualization of system operating status
- Interaction with cloud-based data services

By integrating the ANN model within the Android application, the system achieves intelligent decision-making directly on a mobile platform, enhancing accessibility and response speed.

E. Firebase Realtime Database

Firebase Realtime Database is a cloud-based data synchronization service provided by Google that enables real-time data exchange between connected applications. It allows data to be stored in JSON format and synchronized instantly across multiple clients.

In the proposed system, Firebase Realtime Database is used as the communication interface between MATLAB Simulink and the Android application. MATLAB uploads processed signal data to Firebase in real time, while the Android application retrieves this data for ANN-based classification.

Firebase also supports bidirectional communication, allowing the Android application to upload ANN classification results back to the database. MATLAB retrieves these outputs and stores them in the workspace to support real-time circuit breaker control in the Simulink model.

The use of Firebase enables:

- Low-latency real-time data synchronization
- Cloud-based communication without complex server setup
- Communication-assisted intelligent protection and control

F. Communication and Control Logic

Communication-assisted protection logic is implemented to enable real-time interaction between signal processing, intelligent decision-making, and protective control. The ANN output generated by the Android application is transmitted to Firebase and retrieved by MATLAB in real time.

The retrieved fault classification information is stored in the MATLAB workspace and used as a control signal to operate circuit breakers within the Simulink model. This closed-loop communication framework enables automatic isolation of faulty sections and demonstrates intelligent protection coordination.

This approach simulates the operation of smart grid protection systems by integrating signal processing, artificial intelligence, cloud communication, and automated control within a unified platform.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Simulation Results and Signal Analysis

The MATLAB Simulink model is used to analyze system behavior under normal and faulted operating conditions. Under normal conditions, the three-phase voltage and current waveforms remain balanced and sinusoidal, indicating stable system operation. The negative and zero sequence components are negligible, confirming healthy operating conditions.

During fault conditions, sudden changes in voltage and current magnitudes are observed at the fault inception instant, along with transient disturbances. Unbalanced faults produce noticeable asymmetry among phase quantities, while three-phase faults affect all phases simultaneously. The negative sequence voltage and current components increase significantly during unbalanced faults, clearly distinguishing faulted conditions from normal operation.

Below figure illustrate the contrast between normal and faulted waveforms. These results confirm that the simulated signals accurately capture fault-induced disturbances and provide a reliable foundation for frequency-domain analysis, wavelet-based feature extraction, and ANN-based fault classification.

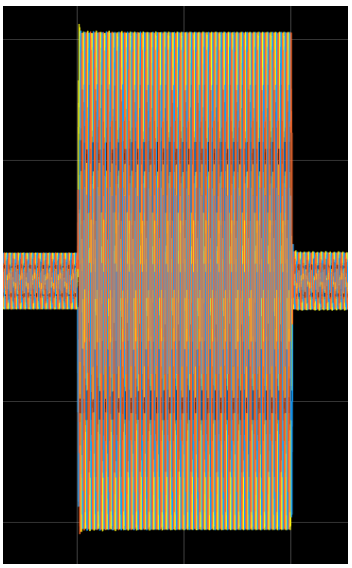

Figure 4.1:ABCG Fault

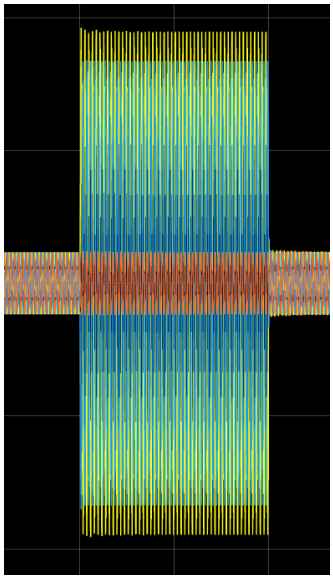

Figure 4.2: AB Fault

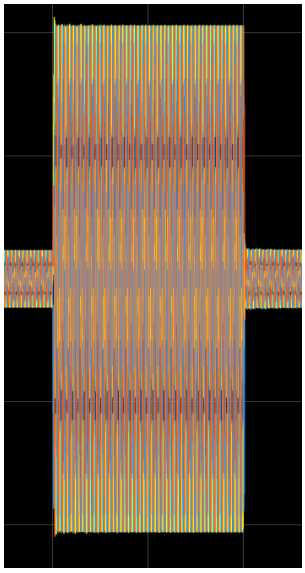

Figure 4.3: ABC Fault

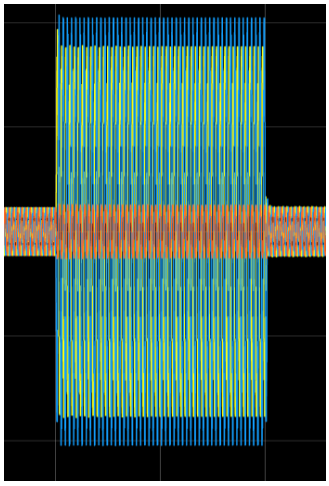

Figure 4.4: ABG Fault

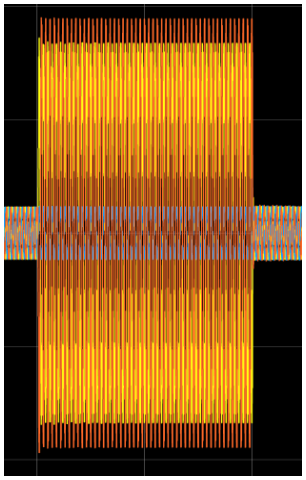

Figure 4.5: AC Fault

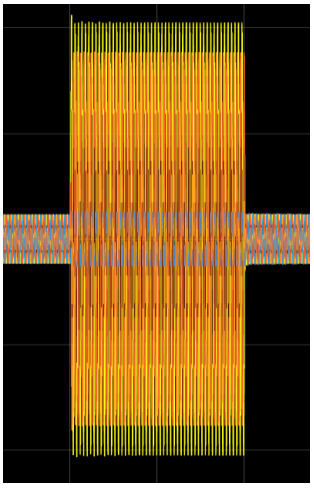

Figure 4.6: ACG Fault

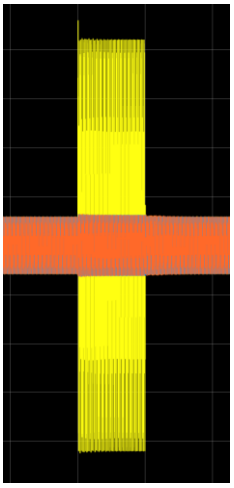

Figure 4.7: AG Fault

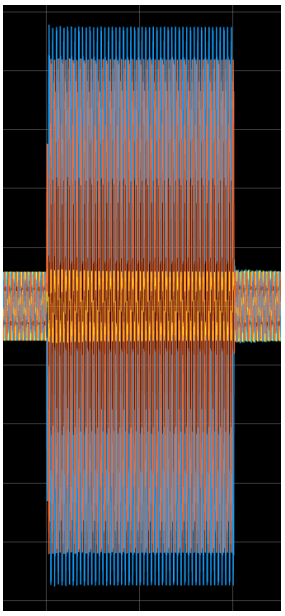

Figure 4.8: BC Fault

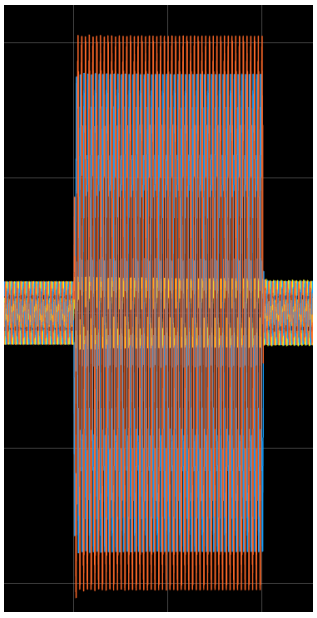

Figure 4.9: BCG Fault

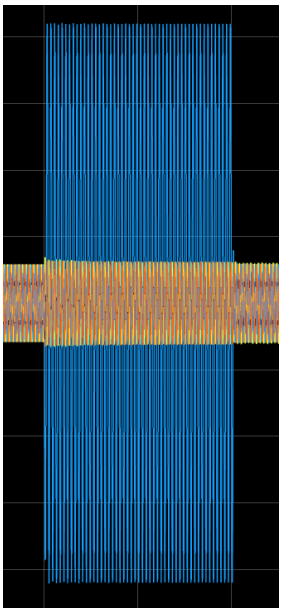

Figure 4.10: BG Fault

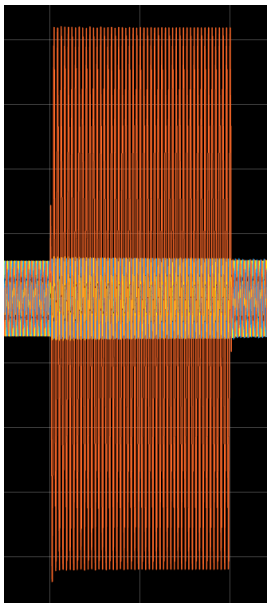

Figure 4.11: CG Fault

3.1 FFT-Based Verification for Wavelet Selection

The FFT analysis is performed on the negative sequence voltage and current components under both normal and faulted operating conditions. Under normal conditions, the negative sequence components exhibit negligible magnitude, resulting in minimal frequency content beyond the fundamental component. This confirms balanced system operation.

During fault conditions, particularly unbalanced faults, the negative sequence voltage and current components show a significant increase in magnitude and the presence of higher-frequency components. The FFT spectrum reveals that fault-induced transients introduce frequency components beyond the fundamental frequency range.

This analysis verifies that the dominant fault-related frequency components present in the negative sequence signals fall within a range effectively captured by the selected Daubechies-4 (db4) wavelet. Hence, FFT analysis on negative sequence quantities is used solely to validate the suitability of the chosen wavelet for transient analysis, rather than for direct fault detection.

Angle	Label	Magnitude	Phase
0	DC	351.5517541	90
10	---	258.9303214	-37.17607566
20	---	209.4457343	244.2281613
30	---	167.2540137	182.2185848
40	---	127.3130137	119.7705219
50	Fnd	100	47.46704897
60	---	87.30373731	-34.23385231
70	---	92.77483631	252.0823269
80	---	92.11725545	188.5538841
90	---	72.64517164	121.7585321
100	h2	60.65681552	44.15359043
110	---	56.16159724	-27.84487542
120	---	53.12598191	257.8281647
130	---	50.84572477	198.3495229
140	---	44.8141954	138.9503407
150	h3	36.72768989	82.50252062
160	---	38.06301342	14.98128842
170	---	48.68785478	-51.7831215
180	---	65.30166046	242.0806974
190	---	78.45065415	172.3135144
200	h4	87.95652911	95.00096505
210	---	94.96853633	13.0471349
220	---	103.5784893	-69.764666
230	---	106.4863936	210.2692347
240	---	100.4963175	134.3692551
250	h5	87.32465334	53.53876329
260	---	81.50370492	-26.07340562

Figure 4.12: Harmonics of Negative Sequence Current

Angle	Label	Magnitude	Phase
0	DC	2188.926955	90
10	---	965.8724918	251.2467216
20	---	820.0770204	209.7484367
30	---	600.7208279	169.8563903
40	---	324.2065465	125.2903693
50	Fnd	100	46.39686499
60	---	169.0164936	-87.1813381
70	---	255.3352922	220.7207797
80	---	252.8917404	173.8311593
90	---	160.1739939	128.9369718
100	h2	60.25451212	51.75961293
110	---	85.77986719	-78.82207825
120	---	143.7053103	233.6995691
130	---	167.9687518	189.6986067
140	---	138.9554023	130.3493489
150	h3	55.3721094	54.31992778
160	---	72.60467578	-61.2956531
170	---	122.7053077	233.3761759
180	---	119.0328018	205.5809935
190	---	102.2327419	133.5201967
200	h4	63.58000432	57.48002271
210	---	68.2660315	-41.42851211
220	---	92.65837511	252.8729319
230	---	104.5835387	197.475195
240	---	90.66343897	135.4531937
250	h5	87.21940572	49.17786428

Figure 4.13: Harmonics of Negative Sequence Voltage

Angle	Label	Magnitude	Phase
0	DC	3383.34118	90
5	---	393.46171	259.08595
10	---	385.291316	248.02538
15	---	370.752548	236.70673
20	---	348.716502	225.0912
25	---	318.097701	213.26811
30	---	278.381619	201.56447
35	---	230.452939	190.79524
40	---	177.936427	182.8452
45	---	129.350923	181.74295
50	Fnd	100	192.17996
55	---	100.336426	205.69485
60	---	115.140121	208.70451
65	---	126.687077	203.60508
70	---	129.147532	195.23743
75	---	122.954662	186.15871
80	---	110.557348	177.68701
85	---	94.6996857	170.7196
90	---	77.8732226	166.17293
95	---	62.2849157	165.22875
100	h2	49.9832069	169.15592
105	---	42.6709452	177.83382
110	---	40.6309222	187.75683
115	---	42.0500035	194.575
120	---	44.4740034	197.22506
125	---	46.275901	196.89454
130	---	46.7721144	195.00268
135	---	45.8960143	192.63401
140	---	43.9347541	190.56205
145	---	41.3542699	189.31495
150	h3	38.6641879	189.16872
155	---	36.2961235	190.09189
160	---	34.5022774	191.73693
165	---	33.3119356	193.57659
170	---	32.5759206	195.14615
175	---	32.0722378	196.21702
180	---	31.6061505	196.80307
185	---	31.0615306	197.0616

Figure 4.14: Harmonics of Negative Sequence Current During Fault

Angle	Label	Magnitude	Phase
0	DC	3007.112087	90
5	---	354.2139339	257.4400132
10	---	340.8456006	244.8845216
15	---	319.177735	232.5925029
20	---	290.3594684	220.7457342
25	---	256.0743968	209.6504144
30	---	218.6912997	199.7862587
35	---	181.3242681	191.834092
40	---	147.5436172	186.5313661
45	---	120.2662189	184.1535044
50	Fnd	100	183.9827694
55	---	84.58204936	185.0489232
60	---	71.83527661	187.7155177
65	---	62.1517682	193.4926677
70	---	57.84646734	202.1709942
75	---	59.68861472	209.8120261
80	---	65.17019865	212.9406169
85	---	70.94120691	211.654848
90	---	74.82929343	207.5130814
95	---	75.9630395	201.9014507
100	h2	74.40360344	194.8824101
105	---	69.29481003	189.7544231
110	---	63.231186	184.796007
115	---	56.23472069	181.235596
120	---	49.27069143	179.6256348
125	---	43.19777292	180.2782715
130	---	38.63148352	182.992521
135	---	35.77302837	186.8736896
140	---	34.35691312	190.677563
145	---	33.83348557	193.504482
150	h3	33.65237699	195.1119875
155	---	33.43125069	195.7153008

Figure 4.15: Harmonics of Negative Sequence Voltage During Fault

3.2 Wavelet-Based Fault Analysis Results

The results of Discrete Wavelet Transform (DWT)–based analysis performed on the negative sequence voltage and current signals obtained from the MATLAB Simulink model. Wavelet analysis is employed to capture transient characteristics associated with different fault conditions, which are not effectively detected using conventional frequency-domain techniques.

The Daubechies-4 (db4) wavelet is used to decompose the negative sequence signals into multiple detail levels. Under normal operating conditions, the wavelet detail coefficients exhibit very low magnitude with no significant transient spikes, indicating a stable and balanced system. This confirms that negative sequence components remain minimal during healthy operation.

During fault conditions, distinct transient patterns are observed in the wavelet detail coefficients. For line-to-ground (LG) faults, a sharp and localized spike appears at the fault inception instant, reflecting the sudden unbalance introduced in the system. Line-to-line (LL) faults result in higher-amplitude wavelet coefficients due to increased fault current magnitude.

In the case of double line-to-ground (LLG) faults, the wavelet coefficients show sustained high-energy transients, indicating a more severe unbalanced disturbance. Three-phase (LLL) faults produce prominent and simultaneous spikes across multiple wavelet detail levels, representing the most severe disturbance among the considered fault types.

Below figure illustrates the wavelet detail coefficients corresponding to different fault conditions. Each fault type exhibits a unique time–frequency signature, enabling clear discrimination between fault categories. These results demonstrate the effectiveness of DWT in isolating fault-related transients and extracting discriminative features for intelligent fault classification.

The wavelet-based analysis confirms that the extracted detail coefficients provide reliable and meaningful inputs for the Artificial Neural Network (ANN), thereby enhancing the accuracy and robustness of the proposed intelligent fault detection and classification system.

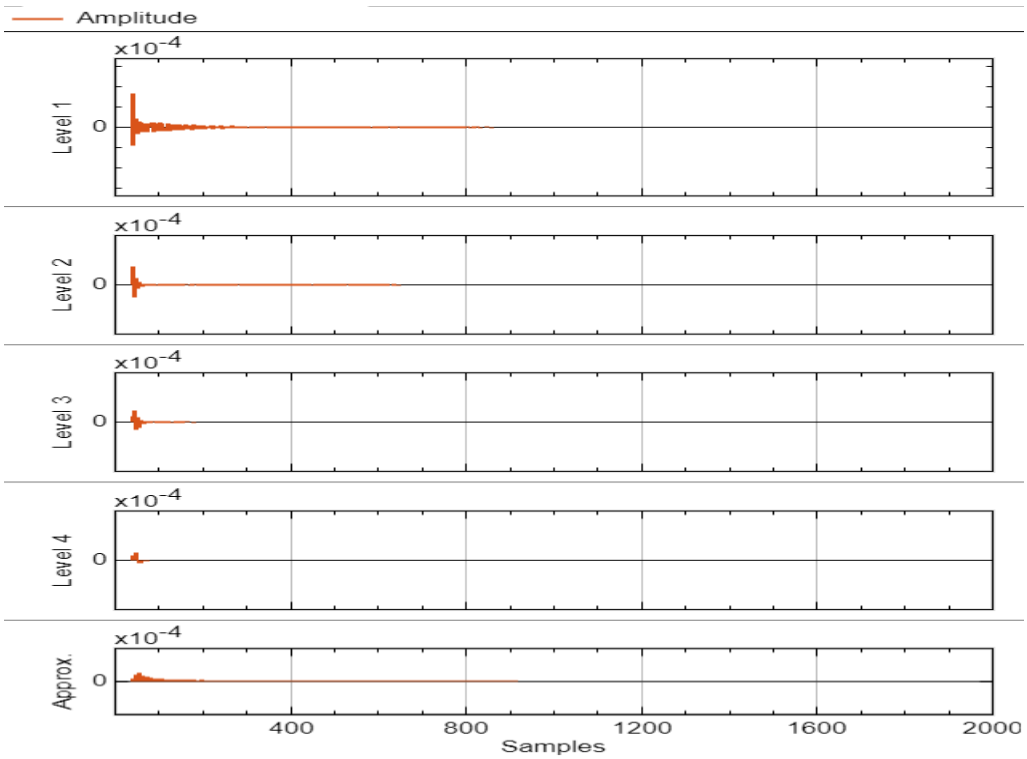

Figure 4.16: Wavelet detail coefficients at Normal Condition

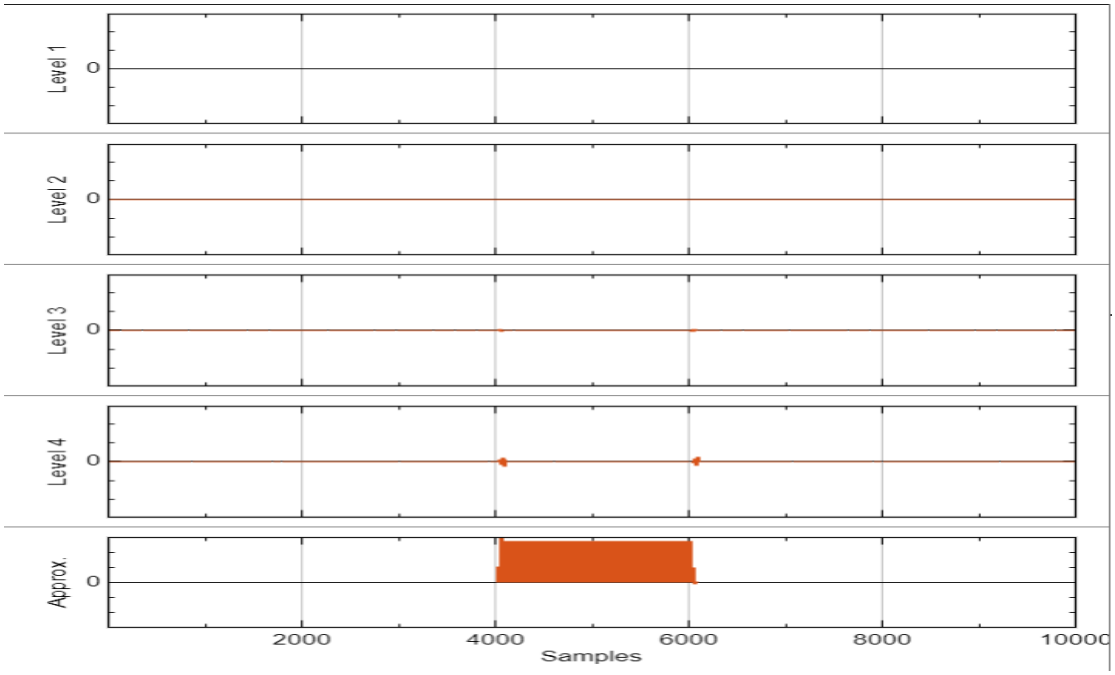

Figure 4.17:Wavelet detail coefficients at LG

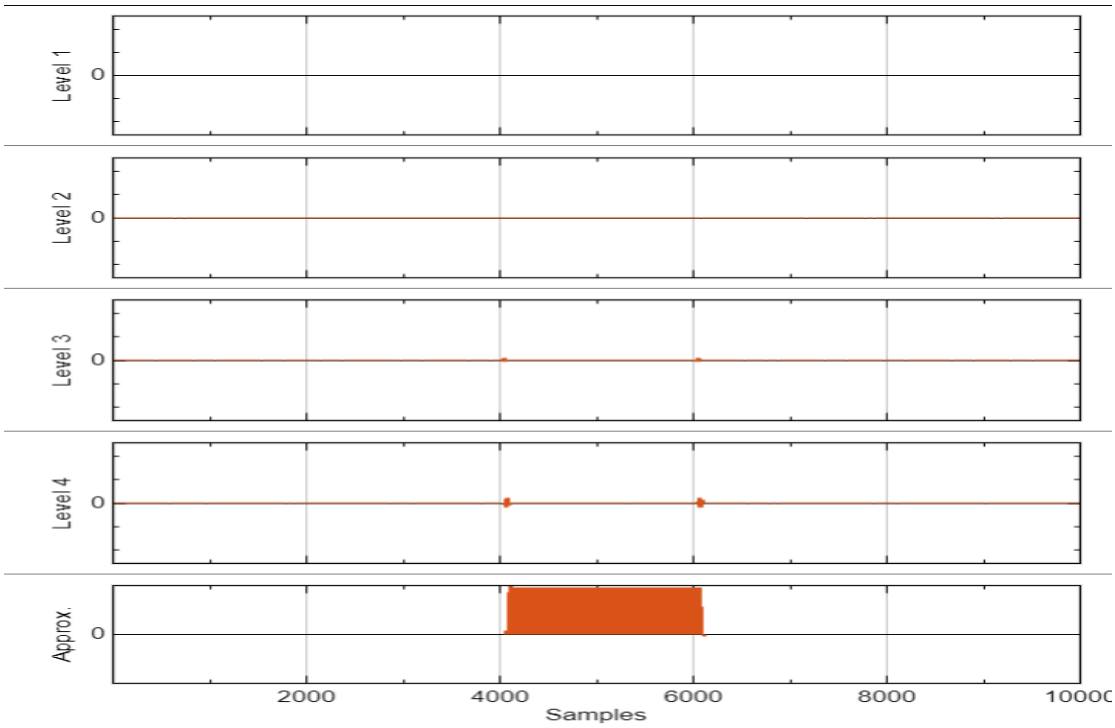

Figure 4.18:Wavelet detail coefficients at LL

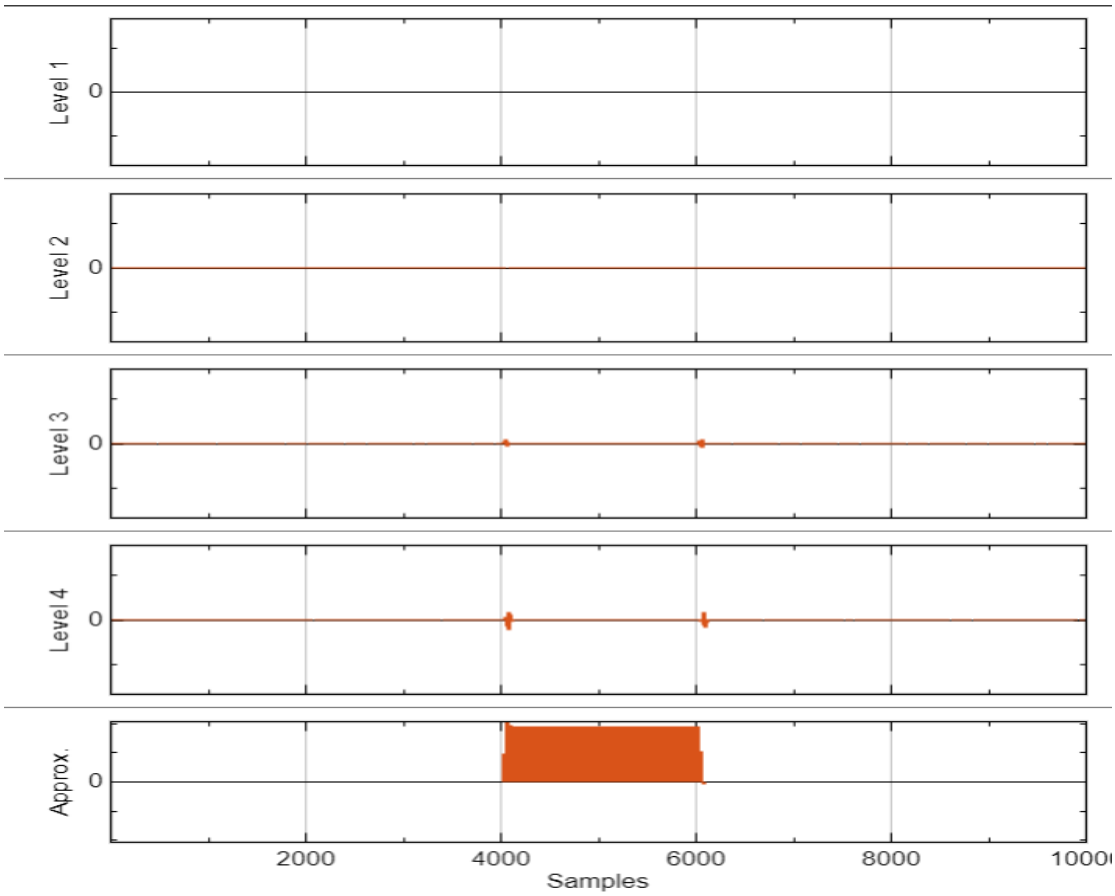

Figure 4.19:Wavelet detail coefficients at LLG

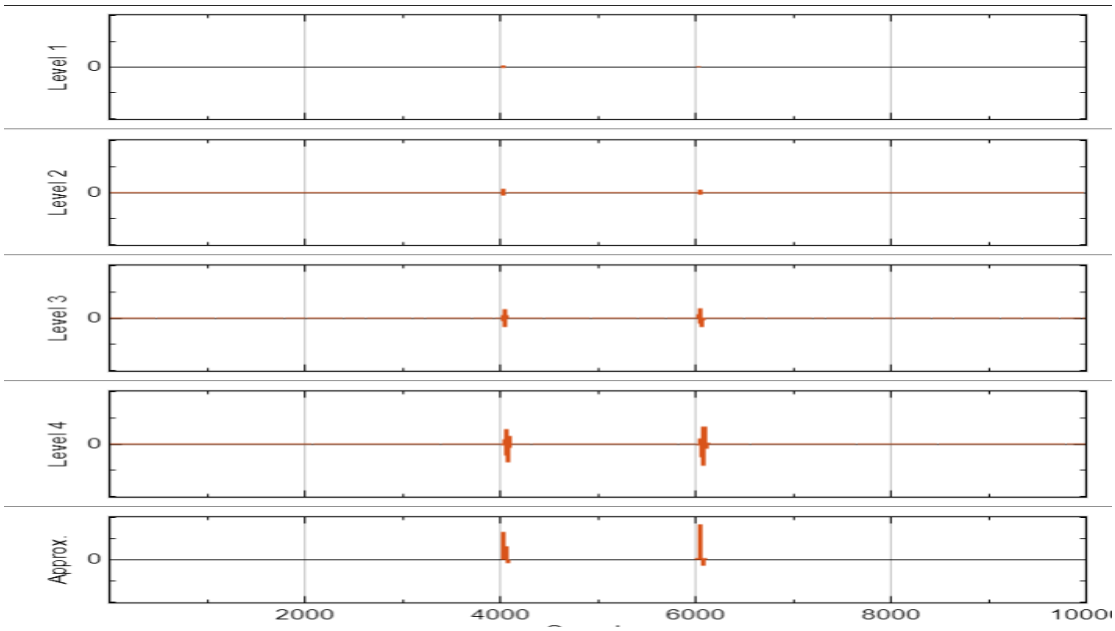

Figure 4.20: Wavelet detail coefficients at LLL

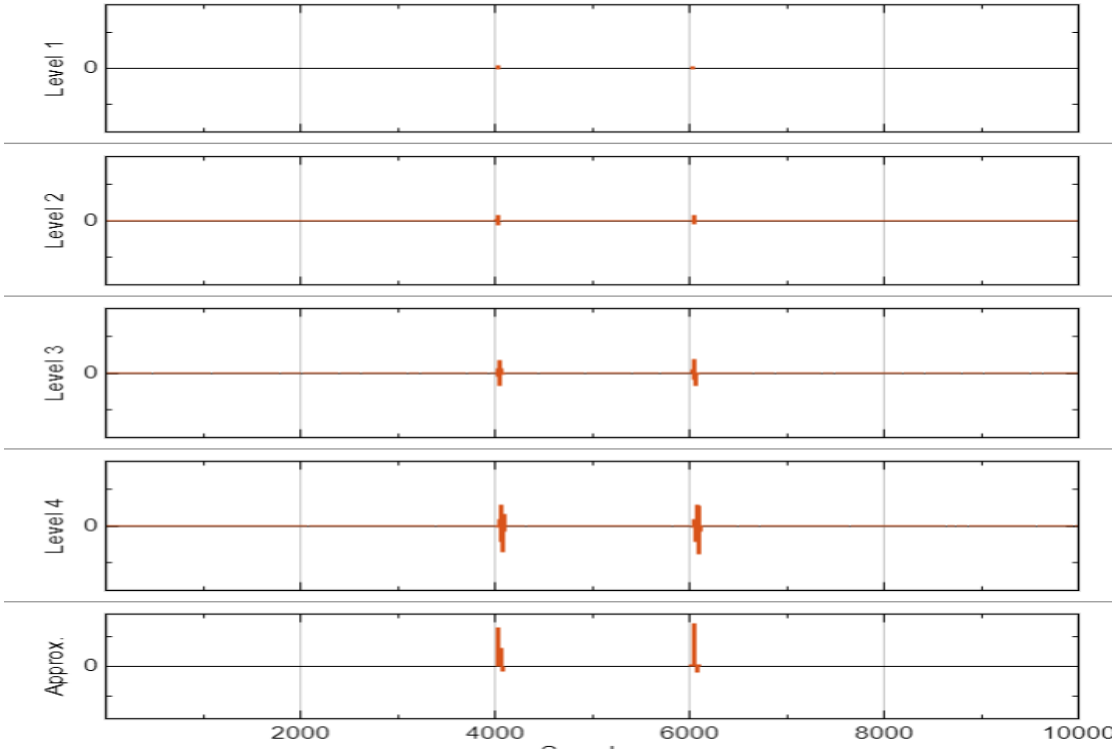

Figure 4.21: Wavelet detail coefficients at LLLG

3.3 ANN-Based Fault Classification Results

The results obtained from the Artificial Neural Network (ANN) employed for fault classification. The ANN performance is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis, to assess its effectiveness under various fault conditions.

3.3.1 Class Distribution Analysis

Below figure illustrates the class distribution of the dataset used for ANN evaluation. Each fault category is represented with nearly equal samples, ensuring balanced learning and unbiased performance assessment. The presence of a separate *No Fault* class further enhances the practical applicability of the model by enabling discrimination between normal and faulted operating conditions.

The balanced dataset contributes significantly to stable training convergence and reliable classification accuracy across all fault types.

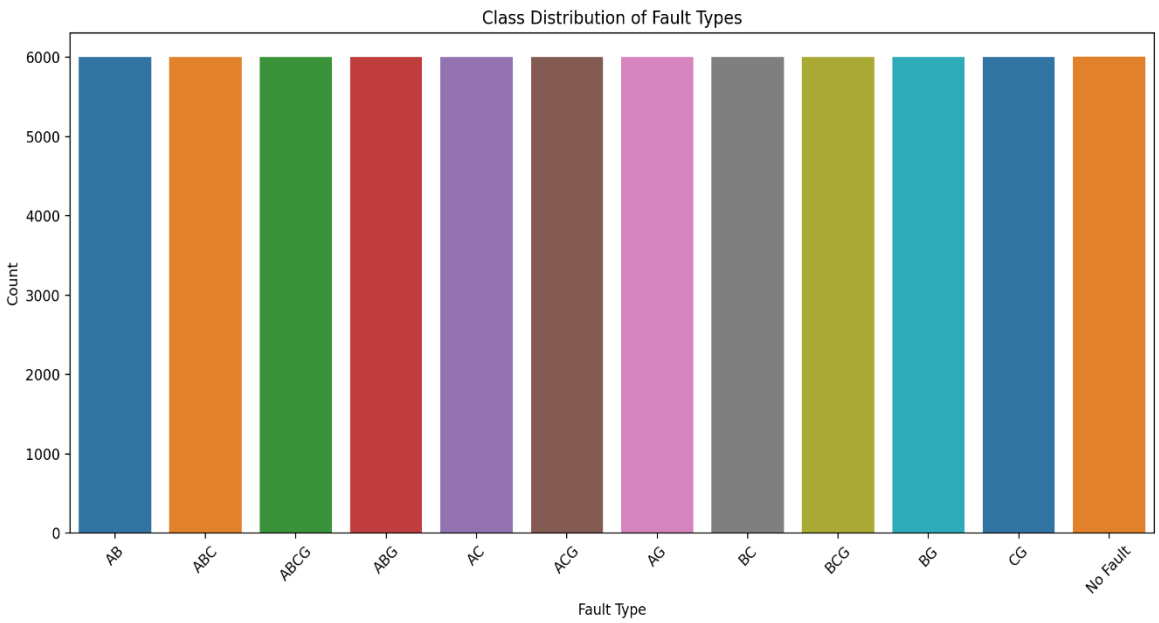

Figure 4.22: Class Distribution of Fault Types

3.3.2 Confusion Matrix Analysis

The confusion matrix shown in Figure 4.23 provides a detailed view of the ANN classification performance. Strong diagonal dominance is observed, indicating that the majority of fault instances are correctly classified. Minimal misclassification occurs primarily between fault types with similar transient characteristics, such as certain phase-to-ground and phase-to-phase faults.

Three-phase and phase-to-phase faults exhibit the highest classification accuracy due to their distinct electrical signatures. The *No Fault* class is identified with near-perfect accuracy, confirming the model's ability to distinguish normal operating conditions from faulted states.

Overall, the confusion matrix confirms the ANN's strong generalization capability and its suitability for real-time fault classification applications.

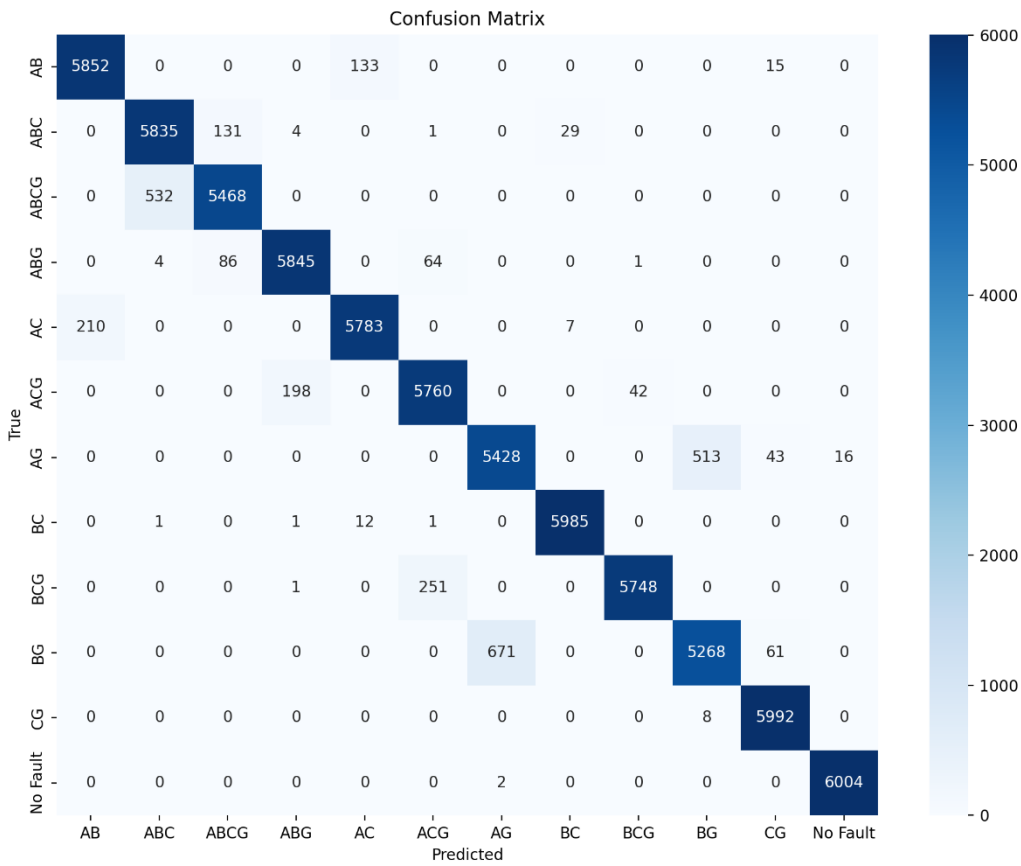

Figure 4.23: Confusion Matrix

4.4.3 Performance Metrics Evaluation

Below figure shows the class-wise performance metrics, including precision, recall, and F1-score. Most fault categories achieve F1-scores exceeding **94%**, demonstrating high classification reliability. Phase-to-phase and three-phase faults achieve the highest performance, while slightly lower metrics are observed for certain single line-to-ground faults due to their relatively lower transient energy content.

Despite this, the ANN maintains consistently high performance across all fault classes, confirming the effectiveness of wavelet-extracted features and sequence component inputs in enhancing fault separability.

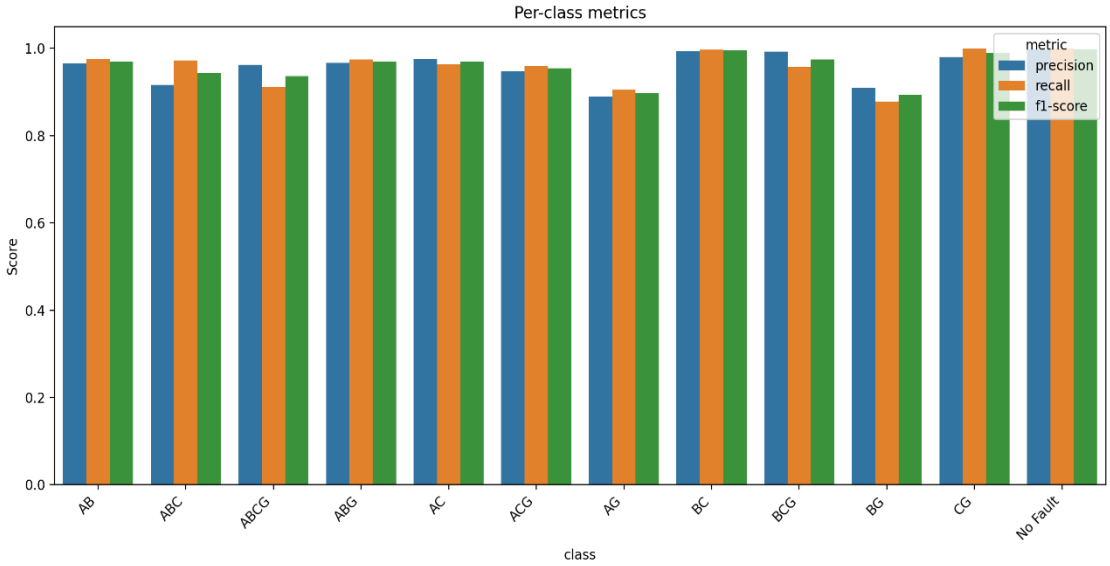

Figure 4.24: Performance Graph

4.4.4 Overall Classification Performance

The trained ANN demonstrates strong fault classification capability across all fault categories. The model achieves an overall classification accuracy of 95.78%, indicating reliable and consistent performance. The macro-averaged precision, recall, and F1-score are 95.81%, 95.78%, and 95.77%, respectively, confirming balanced performance across all classes.

The weighted precision, recall, and F1-score values closely match the macro-averaged metrics, demonstrating that the model performance is not biased toward any specific fault class. These results validate the robustness of the ANN in handling diverse fault scenarios generated from the simulated power system.

4.5 Discussion

The ANN classification results clearly demonstrate the effectiveness of integrating Discrete Wavelet Transform (DWT)-based feature extraction with intelligent learning techniques for power system fault analysis. The high classification accuracy, balanced performance metrics, and strong confusion matrix results validate the robustness of the proposed approach.

The ANN performs reliably under varying fault types and resistances, making it suitable for real-time deployment. These results support the integration of the trained ANN

within the Android application for communication-assisted intelligent protection, as discussed in the subsequent section.

4.5 Android Application and Communication Results

The results related to the development and implementation of the Android application used for real-time monitoring and display of ANN-based fault classification results. The application was developed using Android Studio, which provides a comprehensive environment for Android application development.

The Android application was created using the Empty Views Activity template, as shown in Figure 4.25. The application was developed using the Java programming language, with a minimum supported SDK level of API 24 (Android 7.0 – Nougat). This configuration ensures compatibility with the majority of Android devices while maintaining application stability and performance. The project was configured using the Groovy DSL (build.gradle) build system.

Once the project environment was successfully created, the application was integrated with Firebase Realtime Database to enable real-time data retrieval. The application continuously reads processed data uploaded by MATLAB, including extracted signal features and ANN-based fault classification results.

The received fault classification output is displayed on the mobile interface, allowing users to monitor system status in real time. During testing, the application demonstrated reliable data synchronization with Firebase, confirming low-latency and stable communication between the MATLAB simulation environment and the Android application.

In the final implementation, additional functional sections were incorporated into the Android application to enhance usability and real-time monitoring capability. The application now includes a dedicated Project Information section that provides a summary of the project title, system description, methodology, and technologies used. This section improves user understanding of the overall intelligent fault detection and communication-controlled system directly from the mobile interface.

A Live Monitoring section was also implemented to display real-time sequence voltage and current data retrieved from Firebase. This section continuously updates and presents positive sequence voltage, negative sequence voltage, zero sequence voltage, and negative sequence current values as they are uploaded from MATLAB. The dynamic updating mechanism ensures accurate and continuous monitoring of system electrical parameters during both normal and fault conditions.

Furthermore, a Live Graph section was integrated into the application to provide graphical visualization of sequence voltage and current data in real time. The graph updates dynamically as new data is received from Firebase, allowing users to observe transient behavior and system variations visually. This graphical representation enhances the analytical capability of the application and demonstrates practical deployment potential for smart grid monitoring applications.

In addition to the user interface enhancements, the Firebase Realtime Database structure was organized into multiple nodes to efficiently manage system data. The database includes nodes for sequence voltage data, sequence current data, wavelet-extracted features, ANN classification results, system status, and circuit breaker status. This structured node arrangement ensures clear data organization and reliable synchronization between MATLAB and the Android application.

All relevant application screenshots, including the Project Information interface, Live Monitoring section, Live Graph visualization, and Firebase Realtime Database node structure, are presented below as part of the result validation.

These results validate the successful development of a comprehensive Android-based monitoring interface capable of displaying intelligent fault classification outputs, real-time electrical parameters, and live graphical data. The implemented application demonstrates the feasibility of integrating simulation-based intelligent analysis with mobile platforms and cloud-based communication infrastructure for smart grid monitoring applications.

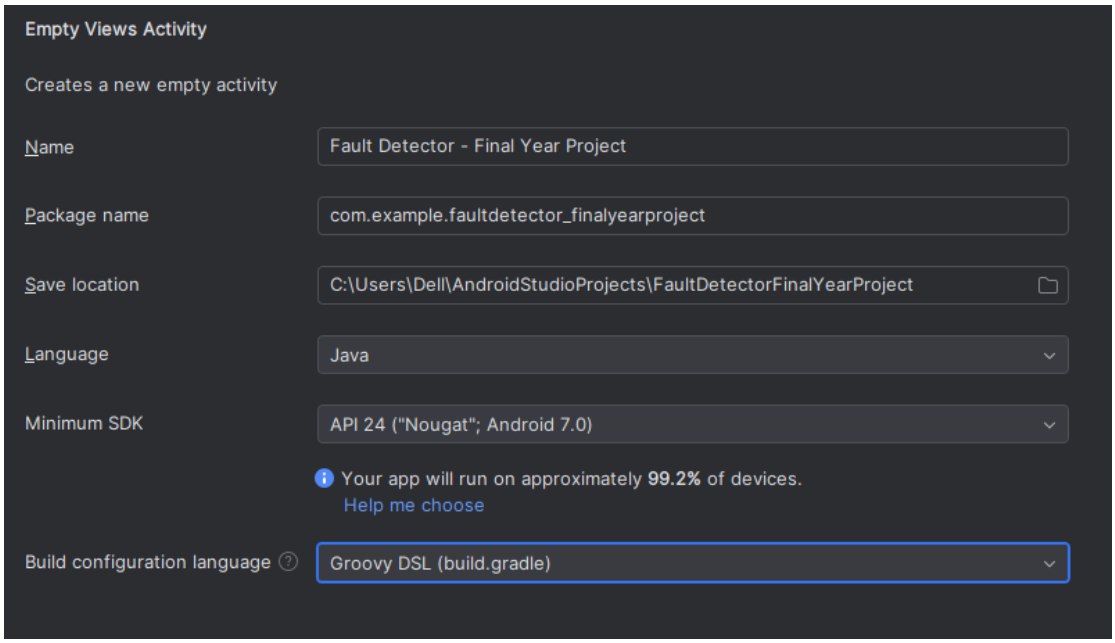

Figure 4.25: App Configuration

Figure 4.26: App Sync Activity

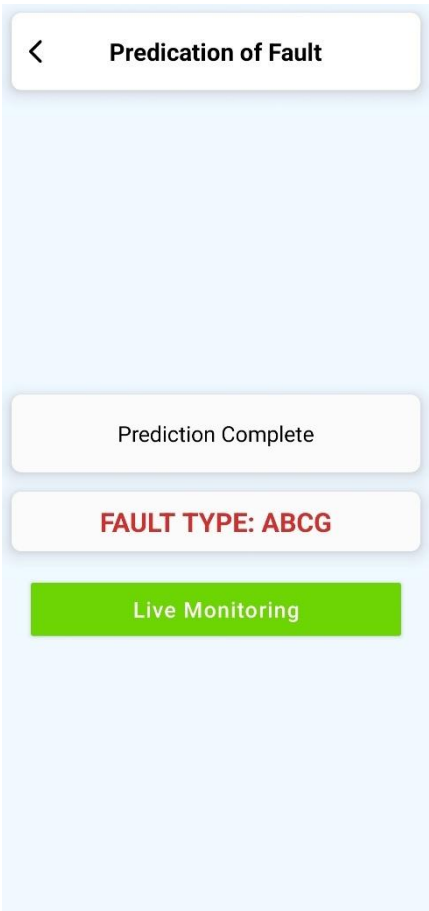

Figure 4.27: App Fault Activity

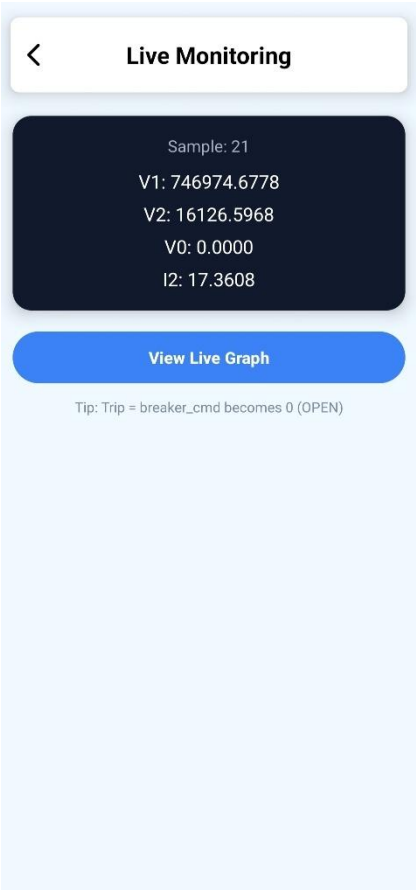

Figure 4.28: Live Monitoring

Figure 4.29: Live Graph

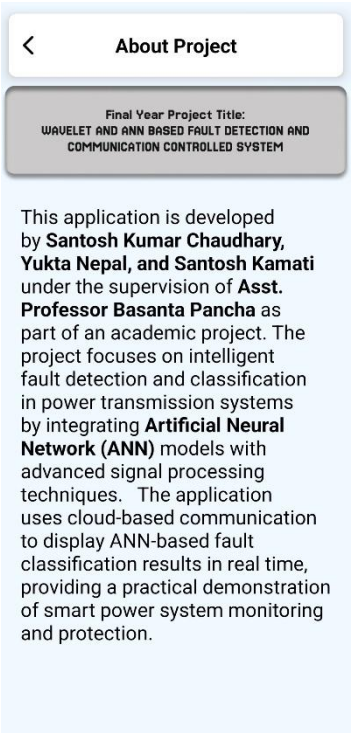

Figure 4.30: About Activity

Figure 4.31: Splash Activity

Figure 4.32: Firebase Realtime Database

4.6 Closed-Loop Circuit Breaker Control and System Validation

This section presents the final implementation and validation of the closed-loop intelligent protection system. The proposed system integrates ANN-based fault classification, Firebase communication, MATLAB workspace monitoring, and automatic circuit breaker operation within the Simulink environment.

The overall Simulink implementation of the closed-loop protection system is shown in the developed model. The model includes sequence analyzers for positive, negative, and zero sequence voltages and currents, three-phase fault blocks, workspace interfacing blocks, and a three-phase circuit breaker controlled through a breaker command signal (breaker_cmd). The ANN classification result transmitted from the Android application via Firebase is continuously monitored by MATLAB and used as a control input to operate the circuit breaker.

When a fault is detected and classified by the ANN, the classification output is sent to Firebase and retrieved in MATLAB. Based on the received fault type, the breaker command signal transitions from closed (1) to open (0), initiating automatic isolation of the faulted section.

The circuit breaker status under closed condition is shown in Figure 4.34, where the control signal remains at logical high state, indicating normal system operation. Under fault detection, the breaker command transitions to open state as shown in Figure 4.33, confirming successful triggering of the protective control logic.

The effectiveness of breaker operation is validated through current and voltage waveform analysis.When the circuit breaker remains closed during fault conditions, high magnitude fault current is observed, as illustrated in Figure 4.35. The waveform shows a significant increase in current amplitude during the fault interval, confirming the presence of a disturbance in the system.

After the ANN-based decision triggers breaker opening, the current waveform reduces to nearly zero, as shown in Figure 4.37. This confirms successful isolation of the faulted section and interruption of fault current flow. The absence of sustained fault current demonstrates correct breaker operation and protective coordination.

Similarly, voltage waveform behavior further validates system performance. With the breaker closed, the system voltage experiences disturbance but continues supplying the faulted section, as observed in Figure 4.36. However, once the breaker opens, the voltage at the isolated section collapses, as shown in Figure 4.38, indicating effective disconnection of the faulted portion from the healthy network.

The transition between closed and open states occurs automatically based on ANN classification without manual intervention. The communication-assisted control loop ensures real-time synchronization between fault detection, classification, and protective action.

The complete operational sequence of the implemented closed-loop system is summarized as follows:

Fault Occurrence →Sequence Component Extraction →Wavelet Feature Extraction →ANN Fault Classification →Firebase Data Synchronization →MATLAB Workspace Update →Breaker Command Generation →Automatic Fault Isolation

Testing was performed under multiple simulated fault conditions, including single line-to-ground, line-to-line, double line-to-ground, and three-phase faults. In all tested cases, the breaker operated correctly based on the ANN output, demonstrating reliable classification-to-control integration.

The results confirm that the proposed intelligent protection framework successfully achieves:

- Real-time ANN-based fault detection and classification
- Cloud-assisted bidirectional communication
- Automatic circuit breaker tripping
- Effective fault isolation
- Stable post-isolation system behavior

This final validation demonstrates the successful implementation of a fully functional communication-assisted, ANN-driven closed-loop protection system. The integration of signal processing, machine learning, cloud synchronization, and automated control confirms the practical feasibility of deploying intelligent smart grid protection systems using simulation and mobile-based monitoring platforms.

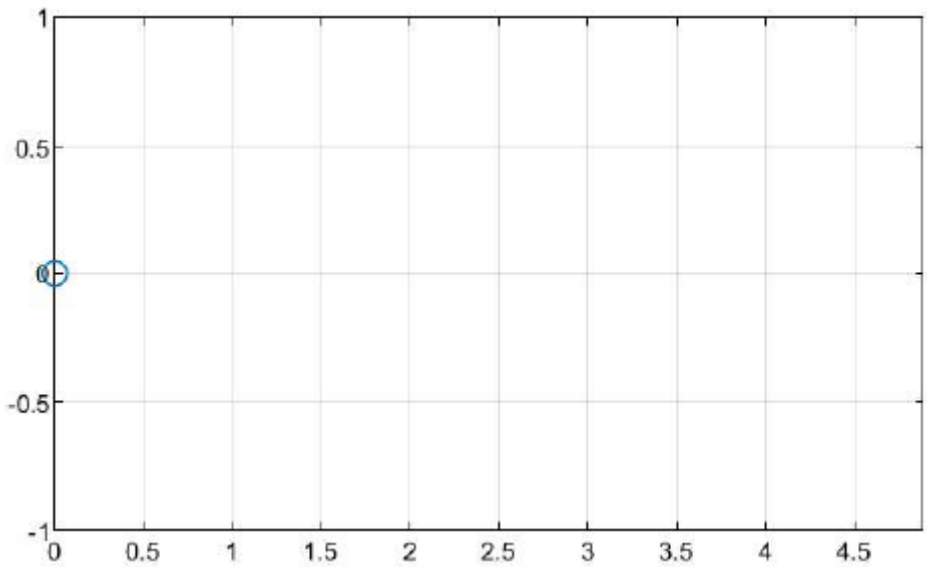

Figure 4.33: Circuit Breaker Open Graph (Value = 0)

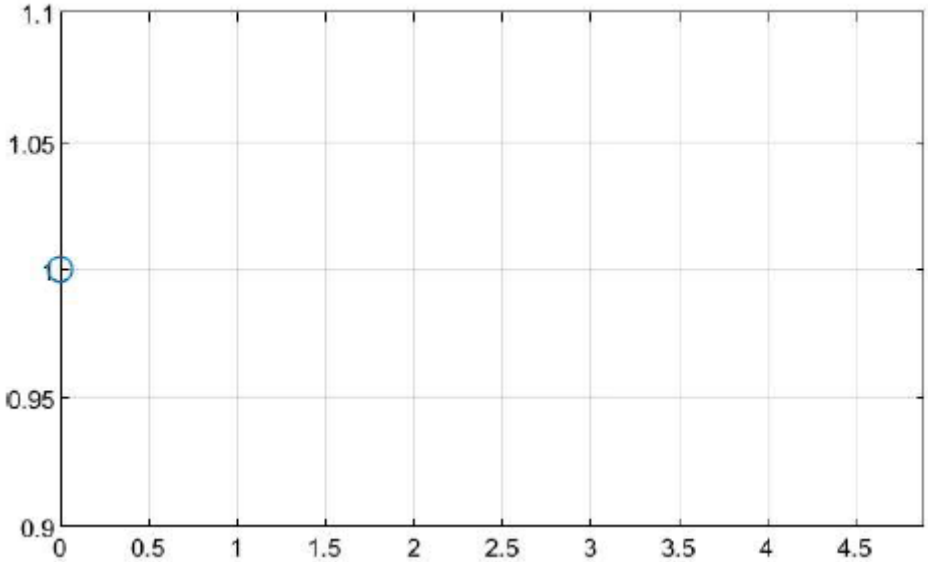

Figure 4.34: Circuit Breaker Closed Graph (Value = 1)

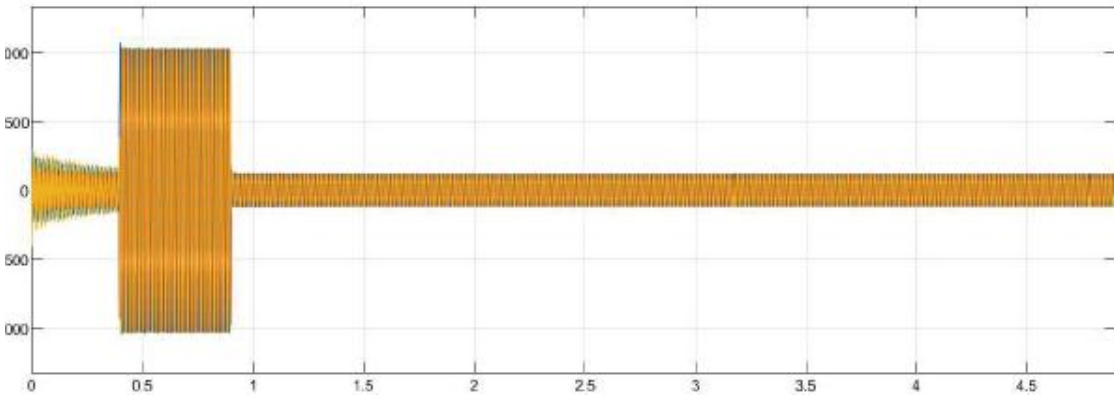

Figure 4.35: Current Waveform During Circuit Breaker Closed

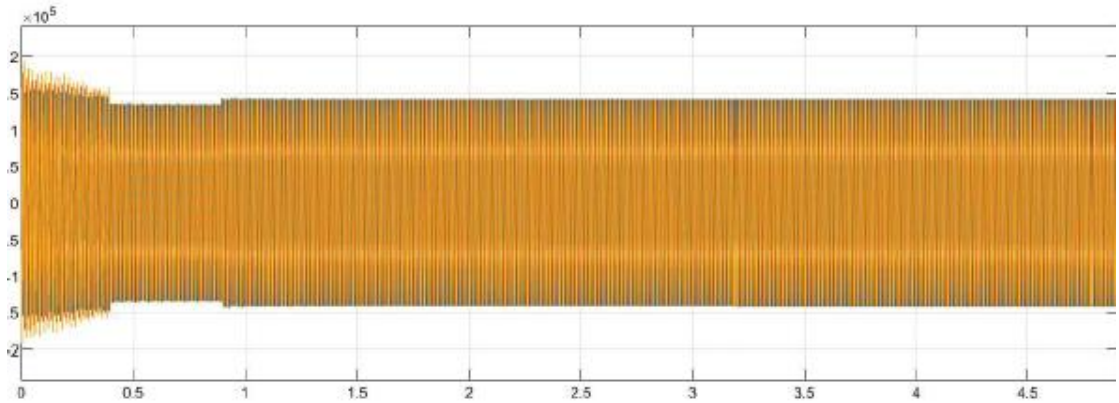

Figure 4.36: Voltage Waveform During Circuit Breaker Closed

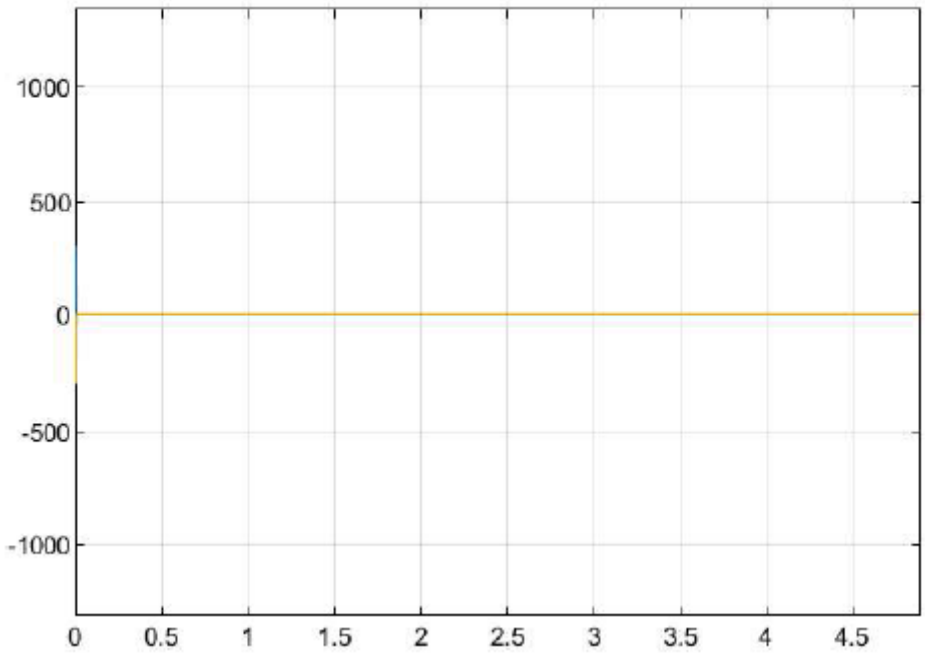

Figure 4.37: Current Waveform During Circuit Breaker Opened

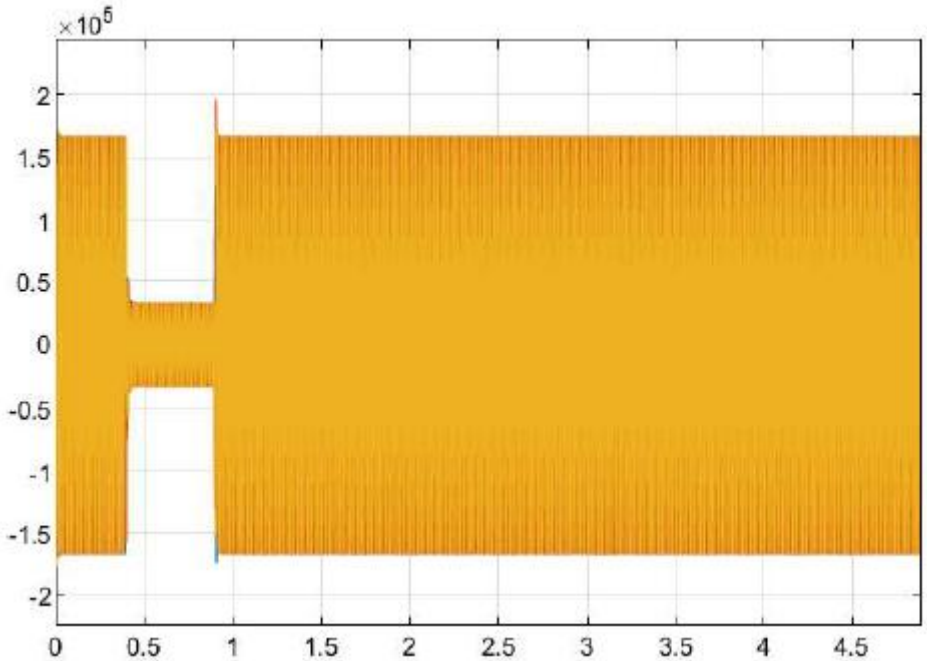

Figure 4.38: Voltage Waveform During Circuit Breaker Closed

CHAPTER 5: CONCLUSION

In this work, a complete intelligent fault detection, classification, and communication-assisted protection system for power transmission lines has been successfully developed and validated using advanced signal processing and machine learning techniques. A detailed power system model was implemented in MATLAB Simulink to generate voltage and current signals under various normal and fault operating conditions. These signals were analyzed to capture system behavior and accurately identify fault-induced disturbances.

Discrete Wavelet Transform (DWT) was applied to extract high-frequency transient features from the simulated signals. The wavelet-based analysis effectively highlighted fault characteristics that are not easily detectable using conventional frequency-domain approaches. The extracted wavelet detail coefficients, along with positive, negative, and zero sequence component information, were used to train an Artificial Neural Network (ANN) for multi-class fault classification.

The trained ANN demonstrated high classification accuracy across multiple fault categories, confirming the effectiveness and robustness of the proposed wavelet–ANN framework. The performance metrics and confusion matrix analysis validated the model's strong generalization capability and reliable discrimination between different fault types.

Beyond fault classification, a fully functional Android application was developed using Android Studio to provide real-time monitoring and visualization of system parameters. The application includes project information, live monitoring of sequence voltages and current, live graphical visualization, and display of ANN-based fault classification results. Firebase Realtime Database was successfully integrated to enable continuous, low-latency, bidirectional communication between MATLAB and the mobile platform.

Most importantly, the system was extended to implement closed-loop protection control. The ANN classification output was transmitted via Firebase and retrieved in MATLAB to generate automatic circuit breaker control signals within the Simulink model. The breaker successfully operated under detected fault conditions, isolating the faulted section and interrupting fault current flow. Waveform analysis

of voltage and current before and after breaker operation confirmed effective fault isolation and stable post-fault system behavior.

The complete operational sequence from fault occurrence, wavelet feature extraction, ANN classification, cloud synchronization, and automatic breaker tripping demonstrates the successful realization of a communication-assisted intelligent protection system.

Overall, the proposed framework effectively integrates power system simulation, signal processing, artificial intelligence, cloud-based communication, mobile monitoring, and automated control. The results confirm the practical feasibility of deploying intelligent, ANN-driven protection systems for modern smart grid applications, contributing toward enhanced reliability, faster fault response, and improved operational efficiency in power transmission networks.

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ANNEX

DATASET FOR FAULT CLASSIFICATION

V1	V2	Vo	Io	D1_v2	D2_v2	D3_v2	D4_v2	D1_i2	D2_i2	D3_i2	D4_i2	Fault Type
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-5.05E-34	-1.35E-07	-1.82E-34	-1.36E-33	-5.05E-34	-4.68E-09	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-5.05E-34	-1.46E-33	-1.82E-34	-1.36E-33	-5.05E-34	-1.46E-33	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-5.05E-34	-2.93E-06	-1.82E-34	-1.36E-33	-5.05E-34	-1.02E-07	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-5.05E-34	-0.002912329	-1.82E-34	-1.36E-33	-5.05E-34	-8.32E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-5.05E-34	-0.077290945	-1.82E-34	-1.36E-33	-5.05E-34	-0.001696569	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	0.000276825	0.07380367	-1.82E-34	-1.36E-33	9.60E-06	0.000766663	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-0.015337121	-0.041904368	-1.82E-34	-1.36E-33	-0.000579596	-0.000116732	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	0.12974185	0.005907562	-1.82E-34	-1.36E-33	0.002404728	0.00012864	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	-0.049655185	-0.001466383	-1.82E-34	-1.36E-33	0.001193248	-1.38E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-1.36E-33	0.004249908	-0.003778942	-1.82E-34	-1.36E-33	0.000918193	2.68E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	0.014887501	0.003411988	0.003532666	-1.82E-34	0.000398235	-5.34E-05	7.65E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	0.104168521	0.009963973	0.000208885	-1.82E-34	0.002579451	-5.60E-05	1.85E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	0.051296766	0.01004094	0.002000258	-1.82E-34	-0.001466458	-7.06E-05	-2.82E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	0.134545103	-0.003006611	-2.56E-05	-1.82E-34	0.00170727	-7.49E-05	-5.12E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-0.030305539	0.00079523	0.000524304	-1.82E-34	-0.000869378	-4.25E-05	6.94E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-0.096560883	0.006745929	-0.002154054	-1.82E-34	0.000665788	9.54E-07	1.73E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-0.104403379	0.012867336	0.002403221	-1.82E-34	-9.73E-06	9.44E-05	-5.59E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	-0.077766479	0.000921175	-0.001251284	-1.82E-34	3.87E-05	7.10E-05	1.18E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	0.043527045	0.006201583	0.000701328	-1.82E-34	0.000164215	1.71E-05	3.44E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-1.82E-34	0.084650134	0.007209989	0.000350864	-1.82E-34	-4.63E-05	-1.46E-05	-1.41E-05	No Fault

1	1.85E-17	7.40E-17	1.85E-17	0.14470714	0.110494007	0.000627612	-0.000240993	0.004133564	8.31E-05	-4.60E-05	-1.88E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-0.138396308	0.054962773	-0.000669389	0.000736325	-0.001402959	-5.55E-05	-5.75E-05	1.83E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	0.00403524	-0.043481229	0.005967433	0.001879569	0.000991081	7.56E-06	3.74E-05	2.15E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	0.011162522	-0.099648036	0.004560746	-0.002700686	-0.000503632	-4.54E-05	-1.82E-05	-4.13E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-0.040098556	-0.10532043	-0.005619112	0.002490764	3.96E-05	-4.53E-05	1.78E-06	-6.18E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-0.015153086	-0.040824275	-0.002481294	-0.001646489	0.00050957	-4.94E-05	3.48E-05	1.28E-05	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-0.003074314	0.052498786	0.001865049	0.001284043	0.000347812	-8.91E-05	-7.67E-06	-2.95E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-0.028176744	0.097089084	-0.002780069	0.00031934	-0.000445017	-4.10E-05	1.46E-05	-3.17E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	0.038965734	0.100948613	-0.005829344	-0.001188924	-0.000483303	-0.000128256	-1.69E-05	2.82E-06	No Fault
1	1.85E-17	7.40E-17	1.85E-17	-0.017198605	0.02608567	-0.000661834	-9.15E-05	0.000252808	-4.45E-05	-3.83E-05	-1.66E-06	No Fault
141531.3531	0.010260921	1.41E-10	0.000125775	-40.73285199	15.34106836	13.06657676	0	-0.003056235	-0.036848821	-0.080188379	0	AG
141524.7802	16.37850138	1.91E-10	0.183774507	-51.17139839	-99.14035188	-83.92315504	0	-0.363612097	1.40024024	-7.933388939	0	AG
141505.7071	107.7127836	3.43E-11	0.859365282	10.64096067	0.785799142	79.9369651	0	-0.095430447	-0.27510217	12.01181667	0	AG
141501.587	96.15589165	1.28E-10	2.185650316	-33.5786649	61.17642987	-90.95530042	0	0.222289679	-2.039157186	-3.502863776	0	AG
141426.8798	231.1777831	2.13E-10	4.550225431	14.84125942	-62.7592532	31.63139417	0	0.15540353	-1.474860739	-5.493635152	0	AG
141383.9499	288.2131905	3.43E-10	8.508873839	-34.55578037	-214.3069728	-33.68703973	0	0.056291721	1.257617549	3.234761915	0	AG
141319.3107	361.821834	3.84E-10	13.4714347	-12.42864152	187.355098	128.7789015	0	-0.16090914	2.050542066	0.497224502	0	AG
141178.6692	509.4598826	4.51E-10	20.22194372	26.28093987	-73.34705401	-2.87065177	0	0.088771594	0.127956955	-2.22712378	0	AG
141152.7969	535.829813	4.45E-10	28.23590696	42.9828352	-22.39178695	-61.65799299	0	-0.074986282	0.33648506	-5.430661638	0	AG
141005.1151	660.8098192	4.21E-10	37.31372652	-12.24774443	-3.81978118	40.52791211	0	-0.08949165	-0.434359383	7.23916895	0	AG
140920.9618	729.5788339	4.90E-10	47.79498916	-21.48301929	32.6814414	4.086322229	0	0.02453154	-1.432311839	-1.404263909	0	AG
140881.1509	766.1856141	4.54E-10	58.74682946	-3.459421843	-90.91428765	-10.94383296	0	0.1744631	1.230946445	-0.573470204	0	AG
140744.4065	856.2762902	6.16E-10	70.44160634	-19.17419794	3.999889956	-2.649650965	0	0.043666558	0.205397301	-0.121206626	0	AG
140763.2968	855.4040382	4.07E-10	82.34662042	-9.873142867	21.04930203	0.933739142	0	-0.077168557	-1.99708386	-0.037572067	0	AG
140695.8235	892.8813938	2.92E-10	93.95248259	34.98182802	-68.45433667	-0.549113485	0	0.02986243	1.05595732	-0.002445718	0	AG

140664.9377	905.4850865	3.78E-10	105.4559088	34.90055876	10.96991464	-2.478587962	0	-0.082192245	0.379806243	-0.004401794	0	AG
140704.8821	900.7944535	5.51E-10	115.97 79292	-20.86311904	6.321011321	-0.392649217	0	-0.06894037	-0.260305514	0.005443579	0	AG
140667.8299	897.0563479	4.77E-10	125.5533406	-28.38340487	17.63834977	1.21911122	0	0.032287194	0.137231535	0.004370704	0	AG
140704.3406	909.1768616	4.63E-10	133.9112466	-3.500580491	4.589915304	-0.364919815	0	0.146925555	-0.052299933	-0.003130753	0	AG
140707.8721	915.8965512	4.89E-10	140.6627297	-0.247569745	0.81685625	-1.337213684	0	0.008363079	-0.052659195	0.004928237	0	AG
140701.1072	925.1671501	6.56E-10	146.0298672	-47.98274179	-3.620814066	0.651356039	0	0.059294291	-0.00138494	-0.00330475	0	AG
140690.2349	1032.615525	5.19E-10	149.7321196	51.06058026	-0.844430622	1.351746184	0	-0.165435677	-0.030927621	-0.008169856	0	AG
140671.4371	1046.2459	4.87E-10	151.997579	18.97470236	-2.207408003	-0.110476715	0	-0.397480776	-0.045370537	0.003433524	0	AG
140608.6221	1183.077944	4.48E-10	153.0532604	-95.73153701	0.381489936	-0.164881626	0	0.211064624	-0.003760009	0.002141452	0	AG
140540.3582	1300.267393	3.76E-10	153.2890348	-41.19499922	5.658020157	1.434132035	0	0.207569909	0.044668797	0.002255453	0	AG
140481.1175	1374.461912	4.49E-10	153.1430172	54.21630864	-2.445244309	1.259402806	0	0.14670056	0.01173265	0.004363669	0	AG
140327.3082	1573.554173	2.04E-10	153.3522293	53.72633413	5.25942602	0.194618199	0	-0.184003431	0.047428595	0.004458683	0	AG
140267.2711	1636.710038	3.77E-10	154.5977176	-17.4311193	-7.529504563	0.872847291	0	-0.259354547	-0.003314597	0.00448586	0	AG
140128.0446	1774.371374	2.80E-10	157.2250745	-8.34657275	2.165822319	1.870172123	0	0.108529211	0.007390381	-2.62E-05	0	AG
139987.8283	1898.837278	2.54E-10	162.2620548	-1.324653084	-8.130156818	0.972637227	0	0.217087145	-0.028680759	0.002178774	0	AG
139377.1129	2514.218089	4.25E-10	273.8812136	-1.569558155	-7.11938061	0	0	0.121579975	0.023798056	0	0	BG
139215.3085	2472.949642	5.93E-10	271.5718328	62.75481985	87.64684644	0	0	0.108365389	-0.565956151	0	0	BG
139023.7376	2556.968897	2.57E-10	266.7182184	-11.5429153	-28.83685412	0	0	-0.256701993	1.204390887	0	0	BG
138882.8437	2530.356622	4.31E-10	259.1916227	-17.74492042	-10.07872843	0	0	-0.135346898	0.270891639	0	0	BG
138687.9941	2675.65585	6.02E-10	249.7804693	14.23996608	-19.53779594	0	0	0.1099999	-0.394425287	0	0	BG
138589.029	2711.098904	7.28E-10	239.6145982	-17.24373914	37.70005037	0	0	-0.002002318	-0.850782956	0	0	BG
138615.7184	2663.986865	4.05E-10	229.6251728	9.699380954	-15.03119483	0	0	-0.169398289	0.001255996	0	0	BG
138719.3078	2551.389984	4.40E-10	220.0179915	3.374839096	0.137046337	0	0	0.05609096	0.293807252	0	0	BG
138808.4225	2470.285637	5.40E-10	210.7338726	28.08355933	-3.816320104	0	0	0.031567349	0.115819815	0	0	BG
138936.3195	2401.62278	2.10E-10	201.9803776	38.19549682	5.751734572	0	0	0.18733184	0.03884715	0	0	BG

139063.093	2340.376539	3.61E-10	193.1711671	24.7468152	3.427051763	0	0	0.143132516	-0.038642974	0	0	BG
139157.6205	2235.064684	4.04E-10	184.8311029	25.93376953	-60.94151419	0	0	-0.08917288	1.034681883	0	0	BG
139307.0646	2226.631016	5.13E-10	176.4910036	-3.917445701	6.01653927	0	0	-0.084629943	-0.498245956	0	0	BG
139319.606	2151.570994	9.68E-10	168.5513686	-50.07359444	35.57142675	0	0	0.040050761	-1.194954177	0	0	BG
139349.0944	2112.70467	4.81E-10	161.6046655	-22.74017553	-9.043052061	0	0	0.139110058	-0.133799763	0	0	BG
139363.809	2094.347678	2.61E-10	155.4332252	28.42900555	-12.84401202	0	0	0.057849825	1.116924366	0	0	BG
139269.2133	2000.031616	3.73E-10	150.0580682	40.90831681	1.451529006	0	0	-0.131004938	0.144331856	0	0	BG
139271.5803	2039.321135	8.65E-10	145.2863473	-5.604767719	-25.81166815	0	0	-0.185264687	-0.150401291	0	0	BG
139159.2456	1992.519088	8.77E-10	141.4018399	-6.471645681	-49.89636394	0	0	-0.060943039	-0.071673667	0	0	BG
139042.9322	1978.958341	1.23E-09	138.1563051	-16.64532157	-1.710103797	0	0	0.091380562	-0.04043016	0	0	BG
138968.9394	1993.179877	8.57E-10	136.0213226	-51.09267281	33.17713435	0	0	0.156035291	-0.055403629	0	0	BG
138823.7004	1956.996926	1.21E-09	134.7900609	39.31936674	48.59023899	0	0	0.078058208	-0.138543788	0	0	BG
138767.0251	1975.894273	1.37E-09	134.5470679	43.30167019	33.5779657	0	0	-0.255955212	-0.113974198	0	0	BG
138699.81	1976.185568	1.34E-09	135.6693426	31.42309875	2.580779481	0	0	-0.185003583	-0.014255746	0	0	BG
138648.5098	2003.263001	1.36E-09	137.450247	-19.7952602	-34.07146431	0	0	0.008064765	0.118207067	0	0	BG
138663.3829	1999.312199	1.20E-09	140.411446	-46.42196725	-44.88008981	0	0	0.052152721	0.125697826	0	0	BG
138703.2925	2051.666206	7.34E-10	144.1237996	11.99156763	-34.22941349	0	0	0.320409456	0.073073404	0	0	BG
138752.7152	2059.686698	7.40E-10	148.6130238	17.95614807	2.280734115	0	0	-0.097646865	0.014721034	0	0	BG
138841.1403	2091.537547	8.09E-10	154.0859226	23.56835505	32.58926933	0	0	-0.283137763	-0.103054019	0	0	BG
138955.6616	2168.880048	6.62E-10	160.2350307	36.95035585	45.7949157	0	0	-0.075478997	-0.159190739	0	0	BG
139374.8959	2514.666308	5.45E-10	273.8542184	-83.67451283	-0.344522578	0	0	0.226406472	-0.000447066	0	0	CG
139222.3346	2649.857049	6.08E-10	272.7183189	19.77599555	0.029195755	0	0	-0.296138526	-0.000221078	0	0	CG
139074.2155	2805.256989	6.31E-10	271.1388745	3.749178177	0.711953539	0	0	-0.346825222	0.001371741	0	0	CG
139026.0121	2906.87282	6.08E-10	269.8170151	5.893122072	0.126301254	0	0	-0.121182861	0.000209839	0	0	CG
138964.967	2956.519079	3.86E-10	269.6489205	-70.22688564	0	0	0	0.232194776	0	0	0	CG

138975.6043	3010.100367	3.67E-10	271.1361985	-0.336011382	0	0	0	0.244108734	0	0	0	CG
138993.5702	2992.45123	5.31E-10	273.7903633	15.93629105	0	0	0	0.018141345	0	0	0	CG
138958.1104	2928.351656	5.69E-10	277.6781764	15.76453157	0	0	0	-0.189818489	0	0	0	CG
139023.4655	2915.7811	7.23E-10	281.8139716	5.489396225	0	0	0	-0.123801511	0	0	0	CG
138960.9209	2790.505601	5.84E-10	285.3060145	-8.925820013	0	0	0	0.155548899	0	0	0	CG
138930.6475	2722.893168	3.50E-10	287.5623217	-12.39864248	0	0	0	0.114268497	0	0	0	CG
138882.4732	2671.324127	7.87E-10	287.8249858	9.80355762	0	0	0	0.032282021	0	0	0	CG
138747.3979	2577.877339	8.39E-10	285.6252688	4.36862259	0	0	0	-0.166027293	0	0	0	CG
138700.0709	2626.416529	8.79E-10	281.0480862	16.48894636	0	0	0	-0.029542959	0	0	0	CG
138532.2509	2608.896194	1.14E-09	274.177561	-10.21697034	0	0	0	0.127335149	0	0	0	CG
138397.8231	2639.878766	1.06E-09	265.0986324	-25.44651114	0	0	0	0.04362079	0	0	0	CG
138274.9391	2738.05458	8.80E-10	254.8537974	-21.92871855	0	0	0	-0.00751827	0	0	0	CG
138118.3076	2794.168212	1.12E-09	243.9897334	33.06980333	0	0	0	-0.033437479	0	0	0	CG
138055.4199	2814.516707	9.38E-10	233.9102292	48.49881987	0	0	0	-0.056376824	0	0	0	CG
138075.9343	2779.853413	1.04E-09	225.0402735	-13.69012591	0	0	0	-0.059004001	0	0	0	CG
138204.4805	2664.413206	1.19E-09	217.1969654	-114.3025518	0	0	0	0.028443578	0	0	0	CG
138269.3364	2611.07237	1.09E-09	210.2762896	-49.17335591	0	0	0	-0.52013354	0	0	0	CG
138399.4533	2561.95755	1.43E-09	204.5704228	26.4656461	0	0	0	-0.022627749	0	0	0	CG
138539.2724	2517.595248	9.29E-10	199.2627298	19.67230123	0	0	0	0.213772639	0	0	0	CG
138610.8835	2450.802557	1.37E-09	194.9461028	-1.2661477	0	0	0	-0.097524983	0	0	0	CG
138782.8495	2461.879964	1.56E-09	191.1424076	25.33624149	0	0	0	0.149405277	0	0	0	CG
138799.8089	2398.982422	1.98E-09	188.0055238	-54.39003209	0	0	0	0.133557142	0	0	0	CG
138846.4631	2385.421667	2.37E-09	185.948938	-67.86611081	0	0	0	-0.007567078	0	0	0	CG
138880.6039	2396.667376	2.37E-09	184.6155459	39.14641966	0	0	0	-0.019582185	0	0	0	CG
138785.2496	2315.153509	2.82E-09	184.1097923	93.89838495	0	0	0	0.021516755	0	0	0	CG

139376.4407	2503.995805	1.11E-09	273.9100053	37.3394757	0	0	0	-0.044891272	0	0	0	AB
139278.6251	2366.974447	1.70E-09	274.2286458	28.86939355	0	0	0	0.232004721	0	0	0	AB
139106.8222	2097.864483	1.99E-09	274.37907	7.853611961	0	0	0	-0.087092418	0	0	0	AB
138975.5184	1975.498752	1.70E-09	273.690952	56.51404599	0	0	0	0.061972026	0	0	0	AB
138755.9789	1718.542895	2.44E-09	271.43307	1.719669104	0	0	0	-0.167405555	0	0	0	AB
138565.8507	1527.467889	2.46E-09	266.621435	-46.91023826	0	0	0	-0.05219245	0	0	0	AB
138422.2063	1390.848303	2.43E-09	259.6426561	-16.52089486	0	0	0	0.120679782	0	0	0	AB
138179.0064	1228.726837	2.44E-09	249.6268852	13.97115595	0	0	0	0.054558862	0	0	0	AB
138118.6095	1146.720269	2.23E-09	236.9827063	-13.42970777	0	0	0	-0.044848688	0	0	0	AB
137969.8829	1112.515798	9.83E-10	222.2390186	-43.64674782	0	0	0	-0.038053157	0	0	0	AB
137888.231	1098.240371	9.99E-10	204.9794021	3.409459465	0	0	0	0.19919892	0	0	0	AB
137919.0883	1069.104214	1.36E-09	186.6400213	10.7448394	0	0	0	0.05923747	0	0	0	AB
137896.4477	1076.266301	1.07E-09	167.3071718	-7.798475943	0	0	0	-0.148973073	0	0	0	AB
137970.8234	1012.441323	1.18E-09	148.0188414	16.90689753	0	0	0	-0.071477577	0	0	0	AB
138147.6201	842.612946	1.48E-09	130.0504224	29.84837753	0	0	0	0.088506955	0	0	0	AB
138277.795	686.6043883	9.70E-10	114.1259104	-23.16288258	0	0	0	0.047341948	0	0	0	AB
138437.8553	494.3636413	1.61E-09	101.0991216	-51.34482185	0	0	0	-0.017276707	0	0	0	AB
138548.0384	336.1254801	1.52E-09	91.19542696	8.809004391	0	0	0	0.047696156	0	0	0	AB
138571.2715	100.6986453	2.08E-09	85.04987896	51.1531511	0	0	0	0.016474197	0	0	0	AB
138656.7743	162.5107053	1.95E-09	83.10056395	6.925104159	0	0	0	-0.128104145	0	0	0	AB
138563.5398	348.3059659	1.66E-09	85.04416091	-48.93848217	0	0	0	0.015324033	0	0	0	AB
138451.7112	591.8471652	1.75E-09	90.73434401	-18.04683603	0	0	0	0.297406252	0	0	0	AB
138340.9581	780.0848514	1.79E-09	99.51396187	-12.16770471	0	0	0	-0.033393321	0	0	0	AB
138082.4913	1057.70081	2.16E-09	111.0176771	-82.51037875	0	0	0	-0.148507764	0	0	0	AB

137926.0266	1235.975994	2.40E-09	125.1651305	10.76400108	0	0	0	-0.184913109	0	0	0	AB
137707.7101	1453.934682	2.34E-09	141.4118956	64.24800515	0	0	0	0.007898624	0	0	0	AB
137483.101	1670.675613	2.12E-09	159.8906629	0.599206437	0	0	0	0.124685516	0	0	0	AB
137361.6465	1790.570693	3.00E-09	179.8671155	-34.01470918	0	0	0	0.03627545	0	0	0	AB
137204.1028	2028.151629	2.28E-09	201.0109622	-11.51529645	0	0	0	-0.016332866	0	0	0	AB
137134.9022	2133.138709	2.07E-09	223.1426067	89.94645155	0	0	0	-0.122749514	0	0	0	AB
138022.3683	4565.624053	3.06E-09	497.484888	-25.18257691	0	0	0	-0.039515964	0	0	0	AC
137935.8039	4428.428258	2.72E-09	495.4065885	-19.50391556	0	0	0	-0.459210454	0	0	0	AC
137994.2798	4288.079333	2.55E-09	491.4151691	5.019728938	0	0	0	0.119950444	0	0	0	AC
138237.6338	3985.700534	3.53E-09	485.4108655	36.18453252	0	0	0	0.131092241	0	0	0	AC
138446.1432	3779.684751	4.05E-09	477.7253467	9.297514193	0	0	0	0.261831748	0	0	0	AC
138654.6632	3571.982622	4.04E-09	468.7057051	-55.81432052	0	0	0	0.050722912	0	0	0	AC
138812.4833	3400.149745	2.67E-09	457.8312174	-32.23472581	0	0	0	-0.035651849	0	0	0	AC
138870.1908	3228.502813	2.17E-09	446.0438902	41.13194093	0	0	0	-0.027350768	0	0	0	AC
138986.5206	3043.83984	1.71E-09	433.1418531	34.7575335	0	0	0	-0.174951042	0	0	0	AC
138901.7361	2892.006542	1.13E-09	419.348405	34.14312131	0	0	0	-0.239867125	0	0	0	AC
138801.4596	2743.990644	1.02E-09	405.7518998	19.62013584	0	0	0	0.288491726	0	0	0	AC
138695.65	2621.393865	1.75E-10	391.9400321	-8.147123527	0	0	0	0.016332022	0	0	0	AC
138452.2698	2454.657733	5.97E-10	378.8145267	-51.59869742	0	0	0	0.046669674	0	0	0	AC
138331.3487	2348.378297	5.67E-10	366.2904535	9.189210487	0	0	0	0.056736654	0	0	0	AC
138136.4792	2187.883298	1.51E-09	354.0105785	35.11963259	0	0	0	-0.035370428	0	0	0	AC
137967.5161	2027.028296	1.08E-09	342.4316763	9.545917827	0	0	0	-0.097235581	0	0	0	AC
137900.8146	1940.49968	9.91E-10	331.1790933	-35.7959995	0	0	0	-0.024649637	0	0	0	AC
137825.418	1763.977579	1.56E-09	321.1509452	-23.84041359	0	0	0	0.172471662	0	0	0	AC
137855.9785	1700.236406	1.95E-09	312.5344416	19.65371128	0	0	0	0.08211671	0	0	0	AC

137919.0671	1599.325969	2.01E-09	304.9431564	41.95663182	0	0	0	-0.083080512	0	0	0	AC
138024.3727	1488.926673	1.94E-09	298.6778376	19.47422833	0	0	0	-0.261462981	0	0	0	AC
138194.4362	1496.372807	2.34E-09	293.7452069	-2.840694134	0	0	0	-0.146058796	0	0	0	AC
138423.7259	1407.598236	2.22E-09	290.6475522	3.327335833	0	0	0	0.356567734	0	0	0	AC
138598.8927	1442.135875	1.55E-09	289.2982787	41.2178919	0	0	0	0.126607326	0	0	0	AC
138769.0996	1480.172005	1.51E-09	289.3760333	-9.575767068	0	0	0	-0.164048957	0	0	0	AC
138950.975	1480.95535	1.50E-09	290.9232779	-8.333315571	0	0	0	-0.165874551	0	0	0	AC
139014.6773	1620.453737	1.95E-09	294.0537388	27.61377433	0	0	0	0.038310514	0	0	0	AC
139124.7688	1659.166835	1.58E-09	298.8657966	-16.01330489	0	0	0	-0.046001967	0	0	0	AC
139093.8474	1788.985606	2.11E-09	305.0228538	-43.06121925	0	0	0	-0.149562731	0	0	0	AC
139000.0946	1942.490002	2.53E-09	312.622721	23.39918316	0	0	0	0.380129759	0	0	0	AC
138017.6967	4566.925131	3.15E-09	497.4604708	-6.44120829	0	0	0	-0.050304547	0	0	0	BC
137701.4433	4246.54438	2.79E-09	496.074854	7.489090135	0	0	0	0.014995831	0	0	0	BC
137357.5433	3938.109448	3.29E-09	492.1732884	5.37812806	0	0	0	0.039294085	0	0	0	BC
137147.1323	3698.451028	3.45E-09	484.9201029	0	0	0	0	0	0	0	0	BC
136928.0876	3545.946097	3.20E-09	474.4853018	0	0	0	0	0	0	0	0	BC
136815.4806	3405.896241	2.44E-09	460.6208202	0	0	0	0	0	0	0	0	BC
136791.0936	3296.524419	2.59E-09	444.6150926	0	0	0	0	0	0	0	0	BC
136823.792	3185.65081	3.32E-09	427.2741556	0	0	0	0	0	0	0	0	BC
137004.2567	3034.650853	3.16E-09	409.8800101	0	0	0	0	0	0	0	0	BC
137263.3539	2751.491451	3.53E-09	393.219951	0	0	0	0	0	0	0	0	BC
137453.5872	2601.308427	3.17E-09	377.3838474	0	0	0	0	0	0	0	0	BC
137671.9792	2387.54361	3.55E-09	362.6565566	0	0	0	0	0	0	0	0	BC
137823.1647	2215.025516	3.54E-09	348.3390465	0	0	0	0	0	0	0	0	BC
137876.345	2065.391401	2.71E-09	335.0937903	0	0	0	0	0	0	0	0	BC

137983.2683	1884.379159	1.77E-09	322.5956537	0	0	0	0	0	0	0	0	BC
137894.5671	1765.029795	2.15E-09	310.8227557	0	0	0	0	0	0	0	0	BC
137819.0741	1617.210703	2.77E-09	300.5651777	0	0	0	0	0	0	0	0	BC
137717.1279	1517.007528	3.22E-09	291.1272456	0	0	0	0	0	0	0	0	BC
137463.0501	1420.642329	3.55E-09	283.4042478	0	0	0	0	0	0	0	0	BC
137341.4002	1321.013226	3.48E-09	277.1598527	0	0	0	0	0	0	0	0	BC
137113.3594	1248.379304	3.42E-09	272.0461998	0	0	0	0	0	0	0	0	BC
136914.2216	1190.130541	3.95E-09	268.8621855	0	0	0	0	0	0	0	0	BC
136826.6934	1212.037917	4.37E-09	266.9704767	0	0	0	0	0	0	0	0	BC
136688.3275	1253.229486	4.54E-09	267.1547445	0	0	0	0	0	0	0	0	BC
136671.3719	1300.248546	3.91E-09	269.2003115	0	0	0	0	0	0	0	0	BC
136697.6155	1389.502898	3.57E-09	272.5044651	0	0	0	0	0	0	0	0	BC
136764.839	1484.038945	3.42E-09	277.7940905	0	0	0	0	0	0	0	0	BC
136922.6929	1634.840857	3.42E-09	284.4372471	0	0	0	0	0	0	0	0	BC
137111.7644	1764.109106	2.43E-09	292.7326141	0	0	0	0	0	0	0	0	BC
137270.9512	1905.373253	2.43E-09	302.4040445	0	0	0	0	0	0	0	0	BC
138011.1807	4558.143015	9.70E-10	497.4984631	0	0	0	0	0	0	0	0	ABC
137910.0305	4609.24658	1.43E-09	498.6441435	0	0	0	0	0	0	0	0	ABC
137764.8365	4570.287125	1.33E-09	501.5345495	0	0	0	0	0	0	0	0	ABC
137648.6675	4586.660078	1.67E-09	506.5156289	0	0	0	0	0	0	0	0	ABC
137453.0196	4432.716435	2.19E-09	513.082656	0	0	0	0	0	0	0	0	ABC
137279.3153	4317.978162	3.08E-09	520.4376039	0	0	0	0	0	0	0	0	ABC
137111.1236	4178.742617	2.46E-09	527.5798737	0	0	0	0	0	0	0	0	ABC
136858.0726	3929.908441	2.35E-09	533.257926	0	0	0	0	0	0	0	0	ABC
136732.722	3810.341594	2.00E-09	536.6451847	0	0	0	0	0	0	0	0	ABC

136511.5896	3591.806078	9.85E-10	536.9727532	0	0	0	0	0	0	0	0	ABC
136353.2877	3437.564663	8.76E-10	533.3476013	0	0	0	0	0	0	0	0	ABC
136264.4145	3362.938578	5.47E-10	525.9828293	0	0	0	0	0	0	0	0	ABC
136088.9036	3251.118861	4.85E-12	514.3985901	0	0	0	0	0	0	0	0	ABC
136063.4517	3258.716539	1.03E-09	498.9972913	0	0	0	0	0	0	0	0	ABC
135971.8346	3276.020289	1.19E-09	480.5744009	0	0	0	0	0	0	0	0	ABC
135908.2633	3320.547828	1.26E-10	459.2208654	0	0	0	0	0	0	0	0	ABC
135896.1392	3396.264647	1.12E-10	436.6468378	0	0	0	0	0	0	0	0	ABC
135861.3968	3455.056882	1.45E-09	413.7615402	0	0	0	0	0	0	0	0	ABC
135832.4067	3499.18545	1.55E-09	391.7568272	0	0	0	0	0	0	0	0	ABC
135825.245	3496.134263	2.23E-09	371.7531938	0	0	0	0	0	0	0	0	ABC
135766.9869	3459.74303	2.50E-09	354.8297742	0	0	0	0	0	0	0	0	ABC
135725.4977	3392.728593	2.62E-09	342.0512595	0	0	0	0	0	0	0	0	ABC
135664.1338	3292.717618	1.98E-09	332.866799	0	0	0	0	0	0	0	0	ABC
135510.6849	3131.984204	1.64E-09	327.6221802	0	0	0	0	0	0	0	0	ABC
135432.4903	2996.967902	9.67E-10	325.2025986	0	0	0	0	0	0	0	0	ABC
135226.1939	2782.427317	4.85E-10	324.5071789	0	0	0	0	0	0	0	0	ABC
135041.1747	2579.116589	4.85E-12	324.8227601	0	0	0	0	0	0	0	0	ABC
134883.3902	2398.918195	4.92E-10	324.6536985	0	0	0	0	0	0	0	0	ABC
134613.7739	2134.563668	4.75E-10	323.4341953	0	0	0	0	0	0	0	0	ABC
134477.4604	1991.200201	2.38E-10	320.4084669	0	0	0	0	0	0	0	0	ABC
134558.2564	3.524287597	5.53E-09	0.024364039	0	0	0	0	0	0	0	0	ABCG
134461.3511	120.9594	6.08E-09	1.31727731	0	0	0	0	0	0	0	0	ABCG
134354.141	209.871887	2.94E-09	4.217087452	0	0	0	0	0	0	0	0	ABCG
134293.8336	358.4221704	3.96E-09	9.024531153	0	0	0	0	0	0	0	0	ABCG

134170.1378	399.870386	4.87E-09	15.30665398	0	0	0	0	0	0	0	0	ABCG
134088.6347	478.533477	3.97E-09	22.45290604	0	0	0	0	0	0	0	0	ABCG
134024.9716	497.9229858	4.72E-09	30.63099183	0	0	0	0	0	0	0	0	ABCG
133982.5205	622.5004028	3.74E-09	39.70650188	0	0	0	0	0	0	0	0	ABCG
133937.083	601.6979948	5.32E-09	49.30092797	0	0	0	0	0	0	0	0	ABCG
133926.0872	650.5028208	4.72E-09	58.72700774	0	0	0	0	0	0	0	0	ABCG
133928.1026	682.4039076	3.70E-09	68.39476204	0	0	0	0	0	0	0	0	ABCG
133882.4118	626.0762575	3.78E-09	77.27625412	0	0	0	0	0	0	0	0	ABCG
133928.1644	677.2896414	3.17E-09	85.51572658	0	0	0	0	0	0	0	0	ABCG
133874.3963	632.7167193	3.61E-09	92.98643759	0	0	0	0	0	0	0	0	ABCG
133853.2006	630.7918726	3.47E-09	98.96405626	0	0	0	0	0	0	0	0	ABCG
133851.4452	655.4129919	3.70E-09	104.0314758	0	0	0	0	0	0	0	0	ABCG
133727.0552	675.1560561	2.78E-09	107.5698154	0	0	0	0	0	0	0	0	ABCG
133672.2701	709.6989541	2.41E-09	110.0830378	0	0	0	0	0	0	0	0	ABCG
133611.2408	803.2868427	2.70E-09	111.5950424	0	0	0	0	0	0	0	0	ABCG
133583.879	787.2524197	2.72E-09	112.1765036	0	0	0	0	0	0	0	0	ABCG
133523.0486	863.0050077	1.63E-09	112.1265153	0	0	0	0	0	0	0	0	ABCG
133500.2356	886.2080628	2.72E-09	112.0334507	0	0	0	0	0	0	0	0	ABCG
133498.1889	896.7601543	1.09E-09	111.2847334	0	0	0	0	0	0	0	0	ABCG
133428.1733	932.8181883	2.39E-09	110.7428694	0	0	0	0	0	0	0	0	ABCG
133479.2122	943.1695679	1.33E-09	110.069069	0	0	0	0	0	0	0	0	ABCG
133426.5865	945.321671	6.72E-10	109.1656483	0	0	0	0	0	0	0	0	ABCG
133420.5179	958.2664706	5.36E-10	108.6899338	0	0	0	0	0	0	0	0	ABCG
133463.1829	978.5463975	1.50E-09	107.9100419	0	0	0	0	0	0	0	0	ABCG
133387.8479	945.8087659	7.95E-10	107.3927885	0	0	0	0	0	0	0	0	ABCG

13345 2.4072	992.34 54868	6.8 6E-12	107.01 13009	0	0	0	0	0	0	0	0	ABCG
13455 7.5434	4.4736 92312	1.5 8E-09	0.0232 19173	0	0	0	0	0	0	0	0	ABG
13452 5.2101	41.295 32673	2.4 3E-09	0.3374 90681	0	0	0	0	0	0	0	0	ABG
13447 5.8075	69.731 24656	1.6 5E-09	1.1984 18414	0	0	0	0	0	0	0	0	ABG
13445 4.5546	134.18 33931	1.2 3E-09	2.8031 94602	0	0	0	0	0	0	0	0	ABG
13435 9.1535	159.62 92273	1.8 8E-09	5.0273 88637	0	0	0	0	0	0	0	0	ABG
13429 4.7951	218.55 02267	1.0 9E-09	7.7311 29803	0	0	0	0	0	0	0	0	ABG
13423 3.6846	244.46 46612	2.9 4E-09	11.255 04634	0	0	0	0	0	0	0	0	ABG
13418 2.1853	350.68 84693	5.2 0E-09	15.711 07278	0	0	0	0	0	0	0	0	ABG
13412 0.8019	350.63 84214	6.4 2E-09	20.911 90324	0	0	0	0	0	0	0	0	ABG
13408 8.9704	412.99 83403	5.2 2E-09	26.394 97265	0	0	0	0	0	0	0	0	ABG
13407 4.5263	465.21 23087	5.3 5E-09	32.650 3647	0	0	0	0	0	0	0	0	ABG
13401 7.0849	435.31 04391	4.8 7E-09	38.843 72359	0	0	0	0	0	0	0	0	ABG
13404 9.3211	521.24 55802	3.5 7E-09	45.832 08409	0	0	0	0	0	0	0	0	ABG
13417 4.5612	652.85 49583	2.2 4E-09	54.706 96921	0	0	0	0	0	0	0	0	ABG
13438 2.6341	916.42 7484	2.4 3E-09	66.456 061	0	0	0	0	0	0	0	0	ABG
13453 6.1774	1068.4 50758	3.3 3E-09	81.026 53723	0	0	0	0	0	0	0	0	ABG
13469 4.791	1220.0 1209	2.2 4E-09	97.328 40286	0	0	0	0	0	0	0	0	ABG
13484 7.921	1356.1 94913	2.9 8E-09	115.31 27904	0	0	0	0	0	0	0	0	ABG
13491 1.7905	1463.6 99912	4.5 7E-09	134.15 82227	0	0	0	0	0	0	0	0	ABG
13504 7.7614	1551.4 67084	3.8 1E-09	153.57 81573	0	0	0	0	0	0	0	0	ABG
13501 7.4676	1609.4 8726	3.0 2E-09	172.82 89947	0	0	0	0	0	0	0	0	ABG
13498 8.5964	1650.1 95366	3.8 6E-09	191.00 96392	0	0	0	0	0	0	0	0	ABG
13495 4.9307	1711.0 97434	3.9 6E-09	207.69 09105	0	0	0	0	0	0	0	0	ABG
13491 6.0396	1669.6 52048	2.9 3E-09	222.28 16878	0	0	0	0	0	0	0	0	ABG

134843.3899	1743.02068	5.34E-09	234.71513	0	0	0	0	0	0	0	0	ABG
134822.6181	1750.364819	3.72E-09	245.1776705	0	0	0	0	0	0	0	0	ABG
134796.1558	1802.904697	5.47E-09	252.8296882	0	0	0	0	0	0	0	0	ABG
134747.3547	1893.44337	4.92E-09	258.4070255	0	0	0	0	0	0	0	0	ABG
134825.7993	2014.164839	6.88E-09	261.5032793	0	0	0	0	0	0	0	0	ABG
134818.858	2147.607509	6.17E-09	262.5766634	0	0	0	0	0	0	0	0	ABG
137073.161	3549.071951	3.07E-09	386.6237904	0	0	0	0	0	0	0	0	ACG
136932.3916	3506.390913	4.45E-09	384.3989014	0	0	0	0	0	0	0	0	ACG
136831.8317	3531.932323	3.30E-09	379.9195515	0	0	0	0	0	0	0	0	ACG
136809.0543	3511.348153	1.75E-09	373.4604206	0	0	0	0	0	0	0	0	ACG
136753.5118	3566.763146	2.59E-09	366.3963365	0	0	0	0	0	0	0	0	ACG
136748.9264	3569.301308	9.94E-10	360.0334805	0	0	0	0	0	0	0	0	ACG
136738.3242	3527.073077	2.33E-10	354.8454498	0	0	0	0	0	0	0	0	ACG
136752.4053	3523.463958	9.70E-10	351.0616767	0	0	0	0	0	0	0	0	ACG
136698.9242	3384.054237	1.21E-09	348.8367872	0	0	0	0	0	0	0	0	ACG
136669.7563	3299.90125	1.08E-11	348.0120714	0	0	0	0	0	0	0	0	ACG
136611.4511	3174.618165	3.70E-09	347.6090448	0	0	0	0	0	0	0	0	ACG
136461.7191	2958.338036	1.20E-09	347.1237149	0	0	0	0	0	0	0	0	ACG
136415.2295	2878.841622	2.04E-09	345.6885275	0	0	0	0	0	0	0	0	ACG
136227.1405	2662.597332	1.55E-09	342.6420753	0	0	0	0	0	0	0	0	ACG
136099.8586	2532.848947	2.21E-09	337.389739	0	0	0	0	0	0	0	0	ACG
135998.9058	2448.412862	9.94E-10	329.8216096	0	0	0	0	0	0	0	0	ACG
135790.9081	2287.51996	1.95E-09	319.4803761	0	0	0	0	0	0	0	0	ACG
135748.0515	2277.681455	7.28E-10	306.562889	0	0	0	0	0	0	0	0	ACG
135644.0362	2220.702128	9.75E-10	291.5297922	0	0	0	0	0	0	0	0	ACG

135659.6117	2170.771979	9.98E-10	275.617155	0	0	0	0	0	0	0	0	ACG
135813.5856	1930.495763	2.31E-09	260.290345	0	0	0	0	0	0	0	0	ACG
135957.4739	1800.577553	9.91E-10	246.0920443	0	0	0	0	0	0	0	0	ACG
136148.9034	1620.319262	1.21E-09	233.0298027	0	0	0	0	0	0	0	0	ACG
136313.4094	1461.31259	2.33E-10	220.7709886	0	0	0	0	0	0	0	0	ACG
136432.9697	1349.977434	2.83E-09	209.4345345	0	0	0	0	0	0	0	0	ACG
136639.8303	1131.408323	4.80E-10	199.2005234	0	0	0	0	0	0	0	0	ACG
136677.822	1041.409191	1.38E-09	189.8597095	0	0	0	0	0	0	0	0	ACG
136707.3333	922.3060758	2.40E-09	181.8332872	0	0	0	0	0	0	0	0	ACG
136725.1274	810.1964155	2.33E-10	174.8550209	0	0	0	0	0	0	0	0	ACG
136592.8494	778.3020849	7.18E-10	169.3259739	0	0	0	0	0	0	0	0	ACG
137058.7664	3538.627885	7.40E-09	386.6313932	0	0	0	0	0	0	0	0	BCG
136820.2418	3300.032227	8.29E-09	385.5864334	0	0	0	0	0	0	0	0	BCG
136536.0436	3049.439763	6.13E-09	382.6657291	0	0	0	0	0	0	0	0	BCG
136347.6323	2838.673949	7.09E-09	376.8536946	0	0	0	0	0	0	0	0	BCG
136135.1211	2726.445959	5.90E-09	367.8689646	0	0	0	0	0	0	0	0	BCG
135989.816	2608.030447	5.17E-09	355.2209434	0	0	0	0	0	0	0	0	BCG
135979.6802	2477.4282	4.94E-09	340.8674968	0	0	0	0	0	0	0	0	BCG
136082.2461	2327.394479	2.39E-09	325.86543	0	0	0	0	0	0	0	0	BCG
136291.7483	2043.85813	1.55E-09	310.9394026	0	0	0	0	0	0	0	0	BCG
136435.2295	1928.927312	2.98E-09	296.3078741	0	0	0	0	0	0	0	0	BCG
136601.8688	1763.852095	3.78E-09	282.0551992	0	0	0	0	0	0	0	0	BCG
136775.5768	1583.365673	2.07E-09	268.1080027	0	0	0	0	0	0	0	0	BCG
136838.846	1466.756617	3.90E-09	255.300298	0	0	0	0	0	0	0	0	BCG
136957.1324	1301.267977	3.91E-09	243.4133461	0	0	0	0	0	0	0	0	BCG

136939.7004	1182.442944	2.92E-09	232.7734343	0	0	0	0	0	0	0	0	BCG
136880.0674	1047.49669	2.99E-09	223.4251096	0	0	0	0	0	0	0	0	BCG
136801.9147	1009.873489	4.99E-09	215.2884267	0	0	0	0	0	0	0	0	BCG
136636.4419	871.4595923	5.53E-09	208.698771	0	0	0	0	0	0	0	0	BCG
136514.1261	831.2149412	6.30E-09	203.7919833	0	0	0	0	0	0	0	0	BCG
136355.6579	788.1537846	4.86E-09	200.1372273	0	0	0	0	0	0	0	0	BCG
136155.5601	736.7228736	3.79E-09	198.2091858	0	0	0	0	0	0	0	0	BCG
136083.7485	826.2117937	3.25E-09	197.7494827	0	0	0	0	0	0	0	0	BCG
135919.5196	848.7912405	3.50E-09	198.8713306	0	0	0	0	0	0	0	0	BCG
135873.1296	936.1807807	1.99E-09	201.5829514	0	0	0	0	0	0	0	0	BCG
135878.7464	1048.590528	3.64E-09	205.5184175	0	0	0	0	0	0	0	0	BCG
135847.1615	1113.650992	4.85E-09	210.8300976	0	0	0	0	0	0	0	0	BCG
135998.5468	1297.037713	5.95E-09	217.3977949	0	0	0	0	0	0	0	0	BCG
136072.7459	1386.039417	5.87E-09	225.1354525	0	0	0	0	0	0	0	0	BCG
136202.3778	1507.28083	6.79E-09	233.7390622	0	0	0	0	0	0	0	0	BCG
136387.6239	1683.141936	7.76E-09	243.1706384	0	0	0	0	0	0	0	0	BCG

Note : There are 72,006 total datasets of every type of fault and no fault , the above dataset contains only first 30 data of each type of fault and no fault.